UM-2 field equations

The coupled field equations of Einstein-Lovelock gravity, the dirac16complex field, the gauge fields G, W^+, W^-, W^3 (Z, γ), B, three generations of quarks and leptons and the tip sigma field, with every interaction, and the flavour evolution of the three neutrinos

Underlying theory: Patrick L. Nash Derived and verified by Claude (Anthropic)

Field equations, version 3, 2026-10-18 (version 2: 2026-10-18; version 1: 2026-10-17)

Repository once-ere/Dirac_claude, folder universe_model_2

Contents

1 How to read this document **2**
1.1 Coordinates, signature, units . . . 2
1.2 Conventions . . . 2

2 The action **4**

3 The gravitational field equations **5**
3.1 The energy-momentum tensors . . . 6

4 The matter field equations **6**
4.1 dirac16complex . . . 6
4.2 The Standard-Model fermions . . . 6
4.3 The gauge fields . . . 7
4.4 The tip sigma field . . . 7
4.5 Boundary conditions . . . 7

5 The background (order 0) **8**
5.1 The metric, the variables and the curvature . . . 8
5.2 The Lovelock tensors of the background . . . 8
5.3 The background field equations . . . 9
5.4 The static block and the stabiliser . . . 10
5.5 The dynamical system, the radion and the response to matter . . . 10
5.6 The four-dimensional effective gravity . . . 10

6 The field equations in the hidden coordinate: the masses **11**
6.1 Gauge fields and the graviton . . . 11
6.2 Fermions . . . 11

7 dirac16complex: the Kohn-Sham equations coupled to all fields **12**
7.1 The portals on the zero modes and the mean fields of an environment (version 3) . . . 12

8 The linear perturbations (order ϵ): CMB, CNB, BAO **15**
8.1 Beyond $n = 0$: the Kaluza-Klein gravitons and the radion (version 3) . . . 15

9 The flavour evolution of the three neutrinos **16**
9.1 The four-fermion interactions at low energy . . . 16
9.2 Vacuum and matter . . . 16

9.3 The thermal potential . . . 17
9.4 The quantum kinetic equations and neutrino decoupling . . . 17
9.5 The dense neutrino gas: the fast flavour instability . . . 18
9.6 The cosmic neutrino background . . . 18

10 The thermal history and the Boltzmann equations (version 3) 19

11 The coupled system and how it is solved 20

12 What was verified 21

1 How to read this document

This document derives and states the complete set of coupled field equations of the universe model UM-2: the gravitational equations with every source, the equations of every matter field with every interaction, their boundary conditions, the exact background (order-0) equations, the exact reductions to the hidden coordinate x_0 that give the particle masses, the Kohn-Sham equations of dirac16complex coupled to all the other fields, and the linear (order ϵ) equations that describe the CMB, the CNB and the BAO, and the equations of the flavour evolution of the three neutrinos in vacuum, in matter, in the early Universe and in dense neutrino gases. It is self-contained: every equation it uses is written here. Labels: AUTHOR (the author's notebooks), RECORD (the Revision record of this repository), STANDARD (textbook), ASSUMED, INVENTED (new in UM-2), DERIVED (computed and verified in UM-2; the verification is stated), ESTIMATE (an order-of-magnitude evaluation), OPEN.

Version 2 (2026-10-18) has three generations of quarks and leptons, anomaly free generation by generation, with 3×3 tip coupling matrices, from which the mixing matrices follow as $U = \kappa_e^\dagger \kappa_\nu$ and $V = \kappa_u^\dagger \kappa_d$ (section 6.2); the flavour evolution of the neutrinos is section 9. The chiral tip Z portal of version 1 is removed (section 2).

Version 3 (2026-10-18) adds the portal I13, an x_0-chiral vector coupling of dirac16complex to the baryon-number current, to the action, the energy-momentum tensor and the field equations (sections 2 to 4); the four-dimensional reduction of the portals and the mean fields of the Standard-Model fields in an environment, which couple the Kohn-Sham equations to every field (section 7.1); the equations of the terms of the metric series beyond $n = 0$ (section 8.1); and the equations of the thermal history: the degrees of freedom, the Boltzmann equations of the dark matter and of the right-handed neutrinos, the freeze-in of the fold bands and the unitarity bounds (section 10). After the independent review of version 3 the I13 sum leaves out the direction x_0, for which the bilinear is not Hermitian (sections 2 to 4); the block response of the coupled solution is the full adiabatic response to every species, the hot plasma included, and the electroweak condensates contain the order-α interaction part (section 7.1); and section 10 adds the self-interaction channel of the dark matter into the fold bands, with the elastic ν_R rate corrected.

1.1 Coordinates, signature, units

The author's coordinates are $X = \{x_0, x_1, \ldots, x_7\}$: x_0 the hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 three extra time-like directions. Frame metric $\eta_{ab} = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ ($a = 0, \ldots, 7$ aligned with $x_0, \ldots, x_7$); capital indices $M, N, \ldots$ run over $x_0, \ldots, x_7$, Greek μ, ν over the four observable coordinates $x_1, \ldots, x_4$ (signature $(+, +, +, -)$). $z = 6Hx_0$; one period of x_0 is $\pi/(6H)$. Units $\hbar = c = 1$; mass dimensions in eight dimensions: $[\Psi] = [\psi] = 7/2$, $[\kappa] = -6$ ($\kappa = 8\pi G_8$).

1.2 Conventions

- The energy-momentum tensor is $T_{MN} = -\frac{2}{\sqrt{|g|}} \frac{\delta S_{matter}}{\delta g^{MN}}$ (vielbein variation for spinors), so that the energy density is $\rho = -T^{x_4}{}_{x_4}$ and the pressures are $p_M = T^M{}_M$ (no sum) for $M \neq x_4$; for a diagonal source $T^M{}_N = \mathrm{diag}(p_0, p_3, p_3, p_3, -\rho, p_t, p_t, p_t)$.
- Gauge fields are in the geometric normalisation: the couplings stand in front of the kinetic terms, $D_M = \partial_M + \Omega_M - iG_M^\alpha T^\alpha - iW_M^a \frac{\tau^a}{2} - iYB_M$, with $[G_M] = [W_M] = [B_M] = 1$ and $[\hat{g}^2] = -4$. Formulas that show $\hat{g}$, $\hat{g}'$, g, g' explicitly refer to the canonically normalised fields $W^a/\hat{g}$, $B/\hat{g}'$, $G^\alpha/\hat{g}_s$, written with the same letters. The weak bosons are $W^\pm = (W^1 \mp iW^2)/\sqrt{2}$: **$W^+$ carries the electric charge $+e$ and W^- the charge $-e$**; $Z = \cos\theta_W W^3 - \sin\theta_W B$, $A = \sin\theta_W W^3 + \cos\theta_W B$.

- The gamma matrices are the author's real 16×16 matrices of $Cl(4,4)$, $\{\gamma^a, \gamma^b\} = 2\eta^{ab}$; $\bar{\Psi} = \Psi^\dagger C$ with $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ (real, symmetric, $C^2 = 1$), $B = -iC\gamma^{x_4}$ (Hermitian, eigenvalues ± 1, eight each), $\Gamma_4 = i\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$, $S^{ab} = \frac{1}{4}[\gamma^a, \gamma^b]$. Since every $C\gamma^a$ is real and antisymmetric, $\bar{\Psi}\gamma^a\Psi$ is anti-Hermitian and the Hermitian vector currents are $i\bar{\Psi}\gamma^a\Psi$; $\bar{\Psi}\Psi$ is Hermitian.
- A delta function on the tip $y = -L$ in proper length is $\delta_{tip} = \delta(y + L)/f$.
- Notation: $\kappa = 8\pi G_8$ throughout (κ_f, with its subscript, is the tip mixing of a fermion). k denotes the block variable $1/f^2$ (k_*, $k_\pm$, δk, $G(k)$); as the index of α_k, $E_{(k)}$ and $\mathcal{L}_{(k)}$ it is the Lovelock order (written p in the reduction formula of section 5.6, where the block variable appears as well); in e^{ikx_1}, in h_j, in $\vec{k}$ and Δk and in the dispersion $\varepsilon = e^{-a_4}|k|$ it is the momentum of a mode along x_1, x_2, x_3; in the perturbation equations of section 8 it is the comoving Fourier wavenumber.
- $\Lambda = -10H^2$ is the eight-dimensional cosmological constant and Λ_4 the four-dimensional one; in the x_0 eigenvalue problems of section 6, $\Lambda = (m\,fL)^2$ is a dimensionless eigenvalue; $\Lambda_\star$ is the scale of the effective interactions; Λ_ℓ and Λ_q are the tip coupling matrices.
- n is a number density (n_o of a Kohn-Sham orbital, n_ψ of a Standard-Model fermion), except as a level number ($n^2\pi^2$, the $n \geq 1$ modes) and in the lattice vector $\vec{n} \in \mathbb{Z}^3$ of a momentum shell. The subscript t (x_t, p_t, v_t) refers to the extra times x_5, x_6, x_7; $\xi = y/L$ is the scaled hidden coordinate; primes denote $d/d\xi$ in section 6 and $d/d\tau$ (conformal time) in section 8 ($\hat{g}'$ and c_Z' are names of couplings). f is the block scale, f_o an occupation number, and the label f in S_f and $T^M{}_N[f]$ stands for the Standard-Model fermions.
- w_k are the Lovelock weights; $w = 1/2$ is the weight of a fold parity in the Kohn-Sham sums, and $w = \sqrt{c^2 - e^2}$ in the fermion mass condition. ψ is a Standard-Model fermion field, except in section 8, where ψ and ϕ are the Newtonian-gauge potentials, and in section 9, where ψ_α are flavour amplitudes. ε is the energy of a mode or orbital.
- Flavour: $i = 1, 2, 3$ is the generation index of the fermion fields and labels the neutrino mass states (i, j are 3-space indices in the background curvatures); $\alpha \in \{e, \mu, \tau\}$ is a flavour index (a colour index on G^α, T^α, $f^{\alpha\beta\gamma}$); U is the PMNS and V the CKM matrix (the gauge transformations are U_L, U_Y, the block basis V_β, the transverse volume $V_\perp$, and the matter potentials carry subscripts, V_{CC}, V_{NC}, $V_{\nu\nu}$); κ_ν, κ_e, κ_u, κ_d are the 3×3 tip matrices; $\rho_{\vec{p}}$ (with a momentum subscript) is a neutrino density matrix; $\mu = \sqrt{2}G_F n_{\nu_e}$ (without subscript) is the unit of the neutrino self-interaction, μ_ψ a bulk mass and μ also the muon; e is the elementary charge, the electron and the fermion eigenvalue εfL; s is also $\sinh w/w$, and $s_{ij} = \sin\theta_{ij}$, $c_{ij} = \cos\theta_{ij}$ (while c with the subscripts L, Q, R, ψ is a localisation $f\mu L$); J is the Jarlskog invariant (and the block operator $\gamma^{x_0}\gamma^{x_1}\gamma^{x_4}$); in section 9 $u = \cos\vartheta$ is the direction cosine of a neutrino (u is also the up quark and $u(y)$ a boson profile).
- Portals: G_V, G_g, G_W, G_B, G_A are four-dimensional coupling strengths (G_N, G_F Newton's and Fermi's constants; G_i the I12 strength of the gauge group i; in section 10 G is the total I12 strength and G_f the coupling of one fermion multiplet; G_A^{B-L} is the strength a $B - L$ variant of I13 would need and G_λ the strength of the self-interaction channel); $\tilde{I}_{of}$, $\tilde{I}_{of}^{(b)}$, $\tilde{I}_\lambda$ overlap integrals; q_ψ the $B - L$ charge of a multiplet and q_f^B its baryon number; $P_a = (1 + \gamma^{x_0})/2$, $P_b = (1 - \gamma^{x_0})/2$ the x_0-chiral projectors ($P_\pm = (1 \pm \Gamma_4)/2$ the four-dimensional ones); J_M^B the baryon-number current. In section 10, $x = M/T$ (M the dark-matter mass, T the temperature), $Y = n/s$ (s the entropy density), g_ρ, g_s effective numbers of degrees of freedom ($g_{s,e\gamma}$ that of the photons and the $e^\pm$ pairs alone), γ a rate density (not a gamma matrix), K_1, K_2 modified Bessel functions of the second kind, N_c the number of Weyl components of a multiplet, which share its overlap (3 for a quark singlet, 6 for a quark doublet, 2 for a lepton doublet, 1 for a charged-lepton singlet), $N_\chi = N_F = 2$ the number of fold bands (the massless fold zero modes of Ψ_A and of Ψ_B, two Dirac fermions) and $\sum_{A,B}$ the sum over the two dark-matter species (the lowest odd states of Ψ_A and of Ψ_B).
- Further symbols. s is the entropy density (section 10), the Mandelstam variable (the squared centre-of-mass energy of a pair, in $\sigma(s)$, $\sqrt{s}$ and the thermal averages), $\sinh w/w$ (section 6.2) and, with indices, $\sin\theta_{ij}$; s_o, s_{DM} and s_{DF} are scalar densities of orbitals. β is the tip parameter of the boson x_0 problems (section 6.1) and, in section 10, the velocity $\sqrt{1 - 4M^2/s}$ of a dark-matter particle in the centre-of-mass frame (V_β is the block basis). p is a pressure (p_0, p_3, p_t, the hidden pressure $\bar{p} = (p_0 + 3p_t)/4$, the order-α pressure p_2 of a gauge plasma), a momentum (of a neutrino in section 9; the centre-of-mass momentum and the integration variable of the ideal-gas integrals in section 10) and the Lovelock order in section 5.6. J is also the total angular momentum of a partial wave (section 10). a is the scale factor e^{a_4} (sections 8 and 10; a_4 is a metric function), $a = 3\eta/2$ in section 8.1, and a, b (with or without a subscript) are the upper and lower components of an orbital. M is a coordinate index, the author's mass parameter in $m = 3HM$, the mass function $M(y)$ of the Kohn-Sham equations (M_{eff} with the mean fields), a static source mass in section

8.1 and the dark-matter mass in section 10 ($M_{Pl} = G_N^{-1/2}$ is the Planck mass). δ marks a variation or perturbation (δk, $\delta \ln f$, $\delta\rho$, $\delta\varepsilon$), the tip delta function δ_{tip} and Kronecker deltas ($\delta_{\alpha e}$, δ_{l0}), and in section 10 $\delta = \frac{1}{3}\, d\ln g_s / d\ln T$. c_ρ, c_3, c_p are the coefficients of the adiabatic block response (section 7.1), b (without a subscript) the one-loop coefficient of the strong coupling and b_i those of $SU(2)$ and $U(1)_Y$.

- H is the author's constant, except in the neutrino-decoupling equations of section 9 and in section 10, where H is the Hubble rate; the time is always x_4 (dots and d/dx_4), and the symmetry axis of the fast-flavour analysis is x_3. Λ^0, Λ^3 (upper indices) are the fast-flavour potentials. $G(k)$ is the Einstein-Hilbert coefficient and $G(u)$ the electron-lepton-number distribution; $F_h = 1.0095$ is the heating factor of ΔN_{eff} and $F(T_1, T_2)$ the energy-transfer kernel.

2 The action

$$S = S_{EL} + S_{stab} + S_\Psi + S_{gauge} + S_f + S_{tip} + S_{I11} + S_{I12} + S_{I13}.$$

Gravity (AUTHOR weights):

$$S_{EL} = \frac{1}{2\kappa}\int d^8x\sqrt{|g|}\Big(\sum_{k=1}^{3}\alpha_k\mathcal{L}_{(k)} - 2\Lambda\Big), \qquad \alpha_k = w_k H^{2-2k}, \quad w = \Big(1, \frac{1}{32}, \frac{1}{1152}\Big), \quad \Lambda = -10H^2,$$

$$\mathcal{L}_{(k)} = \frac{1}{2^k}\,\delta^{A_1B_1\cdots A_kB_k}_{C_1D_1\cdots C_kD_k}\,R^{C_1D_1}{}_{A_1B_1}\cdots R^{C_kD_k}{}_{A_kB_k}: \qquad \mathcal{L}_{(1)} = R, \quad \mathcal{L}_{(2)} = R^2 - 4R_{AB}R^{AB} + R_{ABCD}R^{ABCD}.$$

Block stabiliser (ASSUMED; origin OPEN):

$$S_{stab} = -\int d^8x\sqrt{|g|}\,\rho_b(f), \qquad \kappa\rho_b(f) = \kappa\rho_b^*\Big(\frac{f_*}{f}\Big)^8, \qquad f = e^{-a_4}b_4^{1/2},$$

whose energy-momentum tensor is $T^M{}_N[\mathrm{stab}] = \mathrm{diag}(\rho_b, -\rho_b, -\rho_b, -\rho_b, -\rho_b, \rho_b, \rho_b, \rho_b)$.

dirac16complex (RECORD Lagrangian, canonical spin connection):

$$S_\Psi = \int d^8x\sqrt{|g|}\Big[\tfrac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m\bar\Psi\Psi - \frac{\lambda}{2}(\bar\Psi\Psi)^2\Big], \qquad m = 3HM = H,$$

$$D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \quad D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M, \quad \Omega_M = \tfrac{1}{2}\omega_{Mab}S^{ab},$$
$$\omega_M{}^a{}_b = e^a{}_N\big(\partial_M e_b{}^N + \Gamma^N{}_{ML}e_b{}^L\big), \qquad \gamma^M = e_a{}^M\gamma^a.$$

Gauge fields (ASSUMED structure; $SU(3)_c \times SU(2)_L \times U(1)_Y$ in eight dimensions):

$$S_{gauge} = -\frac{1}{4}\int d^8x\sqrt{|g|}\Big(\frac{1}{\hat g_s^2}G^\alpha_{MN}G^{\alpha MN} + \frac{1}{\hat g^2}W^a_{MN}W^{aMN} + \frac{1}{\hat g'^2}B_{MN}B^{MN}\Big),$$

$$G^\alpha_{MN} = \partial_M G^\alpha_N - \partial_N G^\alpha_M + f^{\alpha\beta\gamma}G^\beta_M G^\gamma_N, \quad W^a_{MN} = \partial_M W^a_N - \partial_N W^a_M + \epsilon^{abc}W^b_M W^c_N, \quad B_{MN} = \partial_M B_N - \partial_N B_M.$$

Fermions (three generations; ASSUMED structure, boundary conditions INVENTED):

$$S_f = \sum_{i=1}^{3}\sum_\psi\int d^8x\sqrt{|g|}\Big[\tfrac{1}{2}\big(\bar\psi_i\gamma^M D_M\psi_i - (D_M\bar\psi_i)\gamma^M\psi_i\big) - \mu_{\psi i}\bar\psi_i\psi_i\Big], \qquad \psi \in \{Q, u_R, d_R, L, e_R, \nu_R\},$$

with, in every generation i, Q_i $(\mathbf{3},\mathbf{2},1/6) = (u,d)_L, (c,s)_L, (t,b)_L$, u_{Ri} $(\mathbf{3},\mathbf{1},2/3) = u_R, c_R, t_R$, d_{Ri} $(\mathbf{3},\mathbf{1},-1/3)$ $= d_R, s_R, b_R$, L_i $(\mathbf{1},\mathbf{2},-1/2) = (\nu_e,e)_L, (\nu_\mu,\mu)_L, (\nu_\tau,\tau)_L$, e_{Ri} $(\mathbf{1},\mathbf{1},-1) = e_R, \mu_R, \tau_R$ and ν_{Ri} $(\mathbf{1},\mathbf{1},0)$; $Q = T_3 + Y$. The bulk masses are diagonal in the generation index, and the doublets of a sector share one bulk mass (ASSUMED): $\mu_{Li} = \mu_L$, $\mu_{Qi} = \mu_Q$.

The tip sigma field (INVENTED; no Higgs particle), on the seven-dimensional tip $y = -L$:

$$S_{tip} = \int_{y=-L} d^7x\sqrt{|g_{tip}|}\Big[-\frac{v_\Sigma^2}{4}\mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu\Sigma\big) - \sum_{i,j}\big(\bar L_i\,\Sigma\,(\Lambda_\ell)_{ij}\,\ell_{Rj} + \bar Q_i\,\Sigma\,(\Lambda_q)_{ij}\,q_{Rj} + \mathrm{h.c.}\big)\Big], \qquad \Sigma \in SU(2),$$

$$D_\mu\Sigma=\partial_\mu\Sigma-iW^a_\mu\frac{\tau^a}{2}\Sigma+iB_\mu\Sigma\frac{\tau^3}{2},\qquad \ell_{Rj}=(\nu_{Rj},e_{Rj})^T,\quad q_{Rj}=(u_{Rj},d_{Rj})^T,$$
$$(\Lambda_\ell)_{ij}=\mathrm{diag}\big((\lambda_\nu)_{ij},(\lambda_e)_{ij}\big),\qquad (\Lambda_q)_{ij}=\mathrm{diag}\big((\lambda_u)_{ij},(\lambda_d)_{ij}\big),$$

with complex 3×3 tip coupling matrices λ_ν, λ_e, λ_u, λ_d; invariant under $\Sigma\to U_L\Sigma U_Y^\dagger$, $U_Y=e^{i\theta\tau^3/2}$, for every choice of the matrices (DERIVED with random group elements and random complex matrices, deviation 1.1×10^{-14}).

Portals (INVENTED):

$$S_{I11}=\int d^8x\sqrt{|g|}\,\frac{c_V}{\Lambda_\star^6}\,(i\bar\Psi\gamma^M\Psi)\,\mathcal{J}_M,\qquad \mathcal{J}_M=\sum_{i=1}^{3}\sum_{\psi}q_\psi\,(i\bar\psi_i\gamma_M\psi_i),$$

with the $B-L$ charges $q_Q=q_{u_R}=q_{d_R}=\frac13$, $q_L=q_{e_R}=q_{\nu_R}=-1$;

$$S_{I12}=-\int d^8x\sqrt{|g|}\,\frac{\bar\Psi\Psi}{\Lambda_\star^3}\,\mathcal{F},\qquad \mathcal{F}=c_GG^\alpha_{MN}G^{\alpha MN}+c_WW^a_{MN}W^{aMN}+c_BB_{MN}B^{MN};$$

$$S_{I13}=\int d^8x\sqrt{|g|}\,\frac{c_A}{\Lambda_\star^6}\sum_{M\neq x_0}(i\bar\Psi\gamma^MP_b\Psi)\,J^B_M,\qquad P_b=\frac{1-\gamma^{x_0}}{2},\qquad J^B_M=\frac13\sum_{i=1}^{3}\sum_{\psi\in\{Q_i,\,u_{Ri},\,d_{Ri}\}}i\bar\psi\gamma_M\psi$$

(version 3; γ^{x_0} the frame matrix; the sum runs over the seven directions $M=x_1,\dots,x_7$; written in the frame it is $\sum_{a=1}^{7}(i\bar\Psi\gamma^aP_b\Psi)\,J^B_a$ with $J^B_a=e_a{}^MJ^B_M$, the same for the diagonal order-0 metric). Since γ^{x_0} anticommutes with every γ^M, $M\neq x_0$, and with C, $P_a\gamma^M=\gamma^MP_b$ and $CP_bC^{-1}=P_a$ ($P_a=(1+\gamma^{x_0})/2$), so $i\bar\Psi\gamma^MP_b\Psi$ is Hermitian for $M\neq x_0$; for $M=x_0$ it is not, $(iC\gamma^{x_0}P_b)^\dagger=iC\gamma^{x_0}P_a$ (it would couple $P_a\Psi$ to $P_b\Psi$ and make the action complex), which is why the sum leaves out x_0 ($J^B_{x_0}$ vanishes on the quark zero modes in any case). The components $M=\mu$ form a four-dimensional vector (P_b commutes with the Lorentz generators $\gamma^\mu\gamma^\nu$); $C\gamma^MP_b$ commutes with B for $M=x_1,\dots,x_4$ and anticommutes for $M=x_5,x_6,x_7$, whose currents vanish for spinors in one half (in particular $J^B_{x_5}$, $J^B_{x_6}$, $J^B_{x_7}$ on the Standard-Model zero modes, which lie in the half $B=-1$), so on the zero modes I13 does not mix Ψ_A and Ψ_B (DERIVED with the author's matrices, exact). In the unitary gauge $\Sigma=1$ the sigma-field kinetic term becomes $-\frac{v_\Sigma^2}{8}\big[2\hat g^2W^+_\mu W^{-\mu}+(\hat g^2+\hat g'^2)Z_\mu Z^\mu\big]$ (canonical fields).

Removed in version 2. Version 1 also contained the chiral tip Z portal $S_{I9}=\int_{y=-L}d^7x\sqrt{|g_{tip}|}\,(c_Z+c'_Z\varsigma)(i\bar\Psi\gamma^\mu P_+\Psi)\,\frac{iv_\Sigma^2}{2}\mathrm{tr}(\Sigma^\dagger D_\mu\Sigma\tau^3)$ ($P_+=(1+\Gamma_4)/2$, $\varsigma=\pm1$ the side of the tip), which let a band of dirac16complex stand in for the neutrinos missing from one generation. With three generations its even part c_Z is bounded by the Z invisible width ($N_\nu=3+2r^2=2.9963\pm0.0074$ gives $r^2<0.0056$; DERIVED) and would still keep two neutrino-like species in equilibrium down to 14.7 MeV, adding

$$\Delta N_{eff}=n_W\Big(\frac{10.75}{g_s}\Big)^{4/3}F_h,\qquad F_h=\Big[\frac{T_\nu}{T_\gamma}\Big(\frac{11}{4}\Big)^{1/3}\Big/(1+h_\nu)\Big]^4=1.0095$$

($n_W=2$ Weyl species decoupling when the plasma has g_s entropy degrees of freedom; F_h corrects for the heating of the neutrinos by e^+e^- annihilation, which an earlier-decoupled species does not receive, with $T_\nu/T_\gamma=0.71664$ and the comoving heating $h_\nu=0.00165$ of the common-temperature decoupling of section 9.4), i.e. $\Delta N_{eff}=2.02$ at $g_s=10.75$; its oriented part c'_Z would thermalise the Ψ_A fold band through $\Psi_A^{odd}\leftrightarrow\Psi_A^{fold}+Z$ and add $\Delta N_{eff}=0.096$ to 0.126 ($g_s=105.75$ to 86.25; ESTIMATE), 2.1 to 2.4 standard deviations above the ACT DR6 value $N_{eff}=2.86\pm0.13$ (disfavoured): UM-2 sets $c_Z=c'_Z=0$ (ASSUMED), so neither fold band of Ψ is ever thermalised and the fold parity is exact in both halves.

3 The gravitational field equations

Varying S with respect to the metric gives

$$\sum_{k=1}^{3}\alpha_k\,E_{(k)}{}^M{}_N+\Lambda\,\delta^M_N=\kappa\,T^M{}_N,\qquad E_{(k)}{}^M{}_N=-\frac{1}{2^{k+1}}\,\delta^{MA_1B_1\cdots A_kB_k}_{NC_1D_1\cdots C_kD_k}\,R^{C_1D_1}{}_{A_1B_1}\cdots R^{C_kD_k}{}_{A_kB_k},$$

$E_{(1)}$ being the Einstein tensor. Every $E_{(k)}$ is divergence-free identically (DERIVED for the order-0 metric for $k=1,2,3$), so the field equations imply $\nabla_MT^M{}_N=0$, which holds on the matter field equations. The total source is

$$T^M{}_N=T^M{}_N[\mathrm{stab}]+T^M{}_N[\Psi]+T^M{}_N[\mathrm{gauge}]+T^M{}_N[f]+T^M{}_N[\mathrm{tip}]+T^M{}_N[I11]+T^M{}_N[I12]+T^M{}_N[I13].$$

3.1 The energy-momentum tensors

Stabiliser. ρ_b depends on the metric through the proper block volume $V_b \propto f^4$, $\rho_b \propto V_b^{-2}$; the variation gives $T_{\mu\nu} = -g_{\mu\nu}\rho_b$ for $\mu,\nu \in \{x_1,\dots,x_4\}$ and $T_{AB} = +g_{AB}\rho_b$ for $A,B \in \{x_0,x_5,x_6,x_7\}$, i.e. $\rho = \rho_b$, $p_3 = -\rho_b$, $p_0 = p_t = \rho_b$.

dirac16complex (and every Standard-Model fermion with $m \to \mu_\psi$, $\lambda \to 0$, D_M with gauge fields):

$$T_{MN}[\Psi] = -\frac{1}{2}\Big[\bar\Psi\gamma_{(M}D_{N)}\Psi - \big(D_{(M}\bar\Psi\big)\gamma_{N)}\Psi\Big] + g_{MN}\,\hat{\mathcal{L}}_\Psi, \qquad \hat{\mathcal{L}}_\Psi = \tfrac{1}{2}\big(\bar\Psi\gamma^P D_P\Psi - (D_P\bar\Psi)\gamma^P\Psi\big) - mS - \frac{\lambda}{2}S^2,$$

with $X_{(MN)} = \frac{1}{2}(X_{MN}+X_{NM})$ and $S = \bar\Psi\Psi$. On shell for a homogeneous state this gives $\rho = mS+U$, $p = SU'-U$ with $U = \frac{\lambda}{2}S^2$ (the record's convention, $\rho = -T^{x_4}{}_{x_4}$).

Gauge fields:

$$\begin{aligned}T_{MN}[\text{gauge}] =& \frac{1}{\hat g_s^2}\Big(G^\alpha_{MP}G^\alpha{}_N{}^P - \frac{1}{4}g_{MN}G^\alpha_{PQ}G^{\alpha PQ}\Big) + \frac{1}{\hat g^2}\Big(W^a_{MP}W^a{}_N{}^P - \frac{1}{4}g_{MN}W^a_{PQ}W^{aPQ}\Big)\\ &+ \frac{1}{\hat g'^2}\Big(B_{MP}B_N{}^P - \frac{1}{4}g_{MN}B_{PQ}B^{PQ}\Big).\end{aligned}$$

Tip (for μ,ν tangent to the tip; zero for the y components):

$$T_{\mu\nu}[\text{tip}] = \delta_{tip}\Big[\frac{v_\Sigma^2}{2}\,\mathrm{tr}\big((D_{(\mu}\Sigma)^\dagger D_{\nu)}\Sigma\big) + g_{\mu\nu}\,\hat{\mathcal{L}}_{tip}\Big], \qquad \hat{\mathcal{L}}_{tip} = -\frac{v_\Sigma^2}{4}\mathrm{tr}\big((D\Sigma)^\dagger D\Sigma\big) - \big(\bar L\Sigma\Lambda_\ell\ell_R + \bar Q\Sigma\Lambda_q q_R + \text{h.c.}\big).$$

Portals. The frame currents $i\bar\Psi\gamma^a\Psi$, $i\bar\psi\gamma^a\psi$ and the scalars $\bar\Psi\Psi$ do not depend on the vielbein, so only $\sqrt{|g|}$ and the explicit metric factors vary:

$$T_{MN}[I11] = g_{MN}\,\frac{c_V}{\Lambda_\star^6}(i\bar\Psi\gamma^P\Psi)\mathcal{J}_P, \qquad T_{MN}[I13] = g_{MN}\,\frac{c_A}{\Lambda_\star^6}\sum_{P\neq x_0}(i\bar\Psi\gamma^P P_b\Psi)J^B_P,$$

$$T_{MN}[I12] = \frac{4\bar\Psi\Psi}{\Lambda_\star^3}\Big[c_G\Big(G^\alpha_{MP}G^\alpha{}_N{}^P - \frac{1}{4}g_{MN}G^\alpha_{PQ}G^{\alpha PQ}\Big) + (W,c_W) + (B,c_B)\Big],$$

so I12 acts as a field-dependent gauge coupling, $1/\hat g_i^2 \to 1/\hat g_i^2 + 4c_i\bar\Psi\Psi/\Lambda_\star^3$.

4 The matter field equations

4.1 dirac16complex

$$\gamma^M D_M\Psi = \Big[m + \lambda\bar\Psi\Psi + \frac{\mathcal{F}}{\Lambda_\star^3}\Big]\Psi - i\,\frac{c_V}{\Lambda_\star^6}\,\mathcal{J}_M\gamma^M\Psi - i\,\frac{c_A}{\Lambda_\star^6}\sum_{M\neq x_0}J^B_M\gamma^M P_b\Psi,$$

and its conjugate. The gravitational term is $\gamma^M\Omega_M$; for the order-0 metric (DERIVED, exact for every a_4, b_4 and on both halves of a period)

$$\gamma^M\Omega_M = \frac{3}{2}\,\frac{H}{f}\,\gamma^{x_0} + \Big(\frac{\dot b_4}{b_4} - \frac{\dot a_4}{2}\Big)\gamma^{x_4},$$

which vanishes only in flat 4+4 space ($H=0$, constant a_4, b_4): the coupling of Ψ to gravity is non-trivial. Ψ is a singlet of the Standard-Model group: it couples to the Standard Model only through I11, I12 and I13.

4.2 The Standard-Model fermions

For a multiplet ψ_i of generation i and charge q_ψ (I11; the I13 term only for the quark multiplets Q_i, u_{Ri}, d_{Ri}),

$$\gamma^M D_M\psi_i = \mu_{\psi i}\,\psi_i - i\,\frac{c_V}{\Lambda_\star^6}\,q_\psi\,(i\bar\Psi\gamma_M\Psi)\,\gamma^M\psi_i - i\,\frac{c_A}{\Lambda_\star^6}\,\frac{1}{3}\sum_{M\neq x_0}(i\bar\Psi\gamma_M P_b\Psi)\,\gamma^M\psi_i + \delta_{tip}\,(\text{Yukawa})_i,$$

$$(\text{Yukawa})_{L_i} = \sum_j \Sigma(\Lambda_\ell)_{ij}\,\ell_{Rj}, \qquad (\text{Yukawa})_{\ell_{Rj}} = \sum_i (\Lambda_\ell)^*_{ij}\Sigma^\dagger L_i,$$

$$(\text{Yukawa})_{Q_i} = \sum_j \Sigma(\Lambda_q)_{ij}\,q_{Rj}, \qquad (\text{Yukawa})_{q_{Rj}} = \sum_i (\Lambda_q)^*_{ij}\Sigma^\dagger Q_i.$$

For one generation the tip terms are equivalent to the self-adjoint mixing conditions at $y=-L$ (DERIVED; block form $\chi=(a,ib)$)

$$b_L(-L) = \kappa_f\, b_R(-L), \qquad a_R(-L) = -\kappa_f\, a_L(-L),$$

with κ_f the dimensionless tip mixing. For three generations they hold in each sector (neutrinos, charged leptons, up-type quarks, down-type quarks) for the vectors of boundary values over the generations,

$$b_L(-L) = \kappa\, b_R(-L), \qquad a_R(-L) = -\kappa^\dagger a_L(-L),$$

with a 3×3 tip matrix κ (κ_ν, κ_e, κ_u, κ_d), and are self-adjoint for every complex κ: the boundary form $\sum[\phi^\dagger\sigma_1\chi-\chi^\dagger\sigma_1\phi]$ of the coupled (L,R) problem over the three generations vanishes identically (DERIVED, sympy, 42 real variables). UM-2 takes (ASSUMED) $\kappa_e=\kappa_u=1$, $\kappa_\nu=U_{PMNS}$, $\kappa_d=V_{CKM}$.

4.3 The gauge fields

With the fermion currents $j_s^{\alpha N}=\sum_q i\bar q\gamma^N T^\alpha q$, $j^{aN}=\sum_{\text{doublets}} i\bar\psi\gamma^N\frac{\tau^a}{2}\psi$, $j_Y^N=\sum_\psi Y_\psi\, i\bar\psi\gamma^N\psi$, the effective inverse couplings $\frac{1}{\hat g^2_{i,eff}}=\frac{1}{\hat g_i^2}+\frac{4c_i\bar\Psi\Psi}{\Lambda_\star^3}$ (I12), and in the unitary gauge $\Sigma=1$ (geometric fields),

$$D_M\Big(\frac{1}{\hat g^2_{s,eff}}G^{\alpha MN}\Big)=j_s^{\alpha N},$$

$$D_M\Big(\frac{1}{\hat g^2_{eff}}W^{1,2\,MN}\Big)=j^{1,2\,N}+\delta_{tip}\frac{v_\Sigma^2}{4}W^{1,2\,N},$$

$$D_M\Big(\frac{1}{\hat g^2_{eff}}W^{3\,MN}\Big)=j^{3N}+\delta_{tip}\frac{v_\Sigma^2}{4}(W^3-B)^N,$$

$$\nabla_M\Big(\frac{1}{\hat g'^2_{eff}}B^{MN}\Big)=j_Y^N-\delta_{tip}\frac{v_\Sigma^2}{4}(W^3-B)^N,$$

where N runs over the tip directions in the δ_{tip} terms and the fermion currents are summed over the three generations. The tip terms are mass terms for $W^\pm$ and Z only: the combination $A=\sin\theta_W W^3+\cos\theta_W B$ (canonical fields) receives none, and the photon stays massless. For an x_0 profile these tip terms become the Robin conditions of section 6.1.

4.4 The tip sigma field

Σ has no radial mode. Its equations of motion are the Noether identities of the gauge symmetry: they follow from the divergence of the tip parts of the gauge-field equations (STANDARD for a gauged non-linear sigma model), so in the unitary gauge $\Sigma=1$ they are not independent; its three modes are the longitudinal W^+, W^- and Z.

4.5 Boundary conditions

The variational principle with the fold reflection $R: z\to\pi-z$ and the regular tip at $y=-L$ gives:

- **gravity and gauge fields**: even (Neumann) or odd (Dirichlet) at the fold, $\partial_y u=0$ or $u=0$; at the tip the natural (Robin) condition from the tip terms;
- **dirac16complex**: fold $(1-\gamma^{x_0})\Psi=0$ (even) or $(1+\gamma^{x_0})\Psi=0$ (odd), both parities being modes of the full period; tip $(1-\gamma^{x_0})\Psi=0$;
- **Standard-Model fermions**, with $P_{sel}=\frac{1+\Gamma_4\gamma^{x_0}}{2}\,\frac{1-i\gamma^{x_5}\gamma^{x_6}}{2}$: doublets ($\mu_\psi>0$) $P_{sel}(1-\gamma^{x_0})\psi=0$ and $(1-P_{sel})(1+\gamma^{x_0})\psi=0$ at the fold, $(1-\gamma^{x_0})\psi=0$ at the tip; singlets ($\mu_\psi<0$) the same with $\gamma^{x_0}\to-\gamma^{x_0}$; at the tip modified by the mixing conditions above.

Each condition is Lagrangian for the boundary form of the Dirac operator, so every operator is self-adjoint (DERIVED).

5 The background (order 0)

5.1 The metric, the variables and the curvature

The order-0 metric is the corrected primordial metric of the Revision record (version 2) in the author's coordinates,

$$\begin{aligned}\bar g = \mathrm{diag}\big(&e^{-2a_4}b_4\cot^2 z,\ e^{2a_4},\ e^{2a_4},\ e^{2a_4},\ -1,\\ &-e^{-2a_4}b_4(\sin^2 z)^{1/6},\ -e^{-2a_4}b_4(\sin^2 z)^{1/6},\ -e^{-2a_4}b_4(\sin^2 z)^{1/6}\big), \qquad z=6Hx_0,\end{aligned}$$

with $a_4(x_4)$, $b_4(x_4)>0$. With $y=\ln|\sin z|/(6H)$ ($dy=\cot z\,dx_0$) it reads

$$ds^2 = e^{2a_4}d\vec{x}^{\,2} - dx_4^2 + f^2\big[dy^2 - e^{2Hy}d\vec{x}_t^{\,2}\big], \qquad f=e^{-a_4}b_4^{1/2}, \quad k=\frac{1}{f^2}, \quad A=\dot a_4, \quad h=\frac{\dot f}{f}=\frac{\dot b_4}{2b_4}-\dot a_4,$$

dots denoting d/dx_4; $\dot k=-2kh$. $\sqrt{|\bar g|}=e^{3a_4}f^4|\cot z|(\sin^2 z)^{1/4}=e^{3a_4}f^4e^{3Hy}|dy/dx_0|$. The Riemann tensor is pair-diagonal ($R^{ab}{}_{cd}=0$ unless $\{a,b\}=\{c,d\}$) and every sectional curvature $K_{ab}=R^{ab}{}_{ab}$ is one of (DERIVED, identical in both charts, independent of x_0)

$$K_{ij}=A^2, \quad K_{i4}=A^2+\dot A, \quad K_{iB}=Ah, \quad K_{4B}=h^2+\dot h, \quad K_{BB'}=h^2-H^2k,$$

$i,j\in\{1,2,3\}$, $B,B'\in\{0,5,6,7\}$. For a pair-diagonal Riemann tensor the Lovelock densities and tensors are (DERIVED; checked against the Ricci route, the explicit Gauss-Bonnet tensor, the Bianchi identities and the minisuperspace variation with a lapse)

$$\mathcal{L}_{(k)}=2^k k!\sum_{k\text{ disjoint pairs}}\prod_{\text{pairs}}K_{ab}, \qquad E_{(k)}{}^{h}{}_{h}=-2^{k-1}k!\sum_{k\text{ disjoint pairs avoiding }h}\prod_{\text{pairs}}K_{ab}.$$

5.2 The Lovelock tensors of the background

Every off-diagonal component vanishes; the x_0 and the extra-time components coincide, $E_{(k)}{}^{x_0}{}_{x_0}=E_{(k)}{}^{x_5}{}_{x_5}$, and nothing depends on x_0 (DERIVED). The independent components are

$$E_{(1)}{}^{x_4}{}_{x_4}=6kH^2-12Ah-6h^2-3A^2$$

$$E_{(1)}{}^{x_1}{}_{x_1}=6kH^2-8Ah-4\dot h-10h^2-2\dot A-3A^2$$

$$E_{(1)}{}^{x_0}{}_{x_0}=3kH^2-9Ah-3\dot h-6h^2-3\dot A-6A^2$$

$$\begin{aligned}E_{(2)}{}^{x_4}{}_{x_4}=&144AhkH^2+24kH^2h^2+72kA^2H^2-12H^4k^2-48hA^3-216A^2h^2\\&-144Ah^3-12h^4\end{aligned}$$

$$\begin{aligned}E_{(2)}{}^{x_1}{}_{x_1}=&96AhkH^2+48\dot hkH^2+72kH^2h^2+48\dot AkH^2+72kA^2H^2-96Ah\dot h\\&-32A\dot Ah-48\dot hh^2-12H^4k^2-48\dot Ah^2-32hA^3-16\dot hA^2\\&-136A^2h^2-192Ah^3-60h^4\end{aligned}$$

$$\begin{aligned}E_{(2)}{}^{x_0}{}_{x_0}=&36AhkH^2+12\dot hkH^2+12kH^2h^2+36\dot AkH^2+72kA^2H^2-72Ah\dot h\\&-72A\dot Ah-12\dot hh^2-36\dot Ah^2-108hA^3-36\dot hA^2-180A^2h^2\\&-12\dot AA^2-108Ah^3-12h^4-12A^4\end{aligned}$$

$$E_{(3)}{}^{x_4}{}_{x_4}=864hkA^3H^2+1296kA^2H^2h^2-216A^2H^4k^2-1440A^3h^3-1080A^2h^4$$

$$\begin{aligned}E_{(3)}{}^{x_1}{}_{x_1}=&576Ah\dot hkH^2+576A\dot AhkH^2+288\dot AkH^2h^2+576hkA^3H^2+288\dot hkA^2H^2+1008kA^2H^2h^2\\&+576AkH^2h^3-144\dot AH^4k^2-864\dot hA^2h^2-216A^2H^4k^2-576A\dot hh^3-576A\dot Ah^3\\&-144\dot Ah^4-576A^3h^3-1368A^2h^4-576Ah^5\end{aligned}$$

$$\begin{aligned}E_{(3)}{}^{x_0}{}_{x_0}=&432A\dot AhkH^2+648hkA^3H^2+216\dot hkA^2H^2+216kA^2H^2h^2+216\dot AkA^2H^2+216kA^4H^2\\&-432h\dot hA^3-648\dot hA^2h^2-648\dot AA^2h^2-432A\dot Ah^3-648A^4h^2-1224A^3h^3\\&-648A^2h^4\end{aligned}$$

5.3 The background field equations

With the author's weights and $\Lambda = -10H^2$ the three independent components are (DERIVED; the x_4 component is a constraint, the other two are linear in $\dot{A}$ and $\dot{h}$)

$$
\begin{aligned}
x_4: \quad & -\frac{15A^2h^4}{16H^4} - \frac{5A^3h^3}{4H^4} - \frac{3h^4}{8H^2} - \frac{9Ah^3}{2H^2} - \frac{27A^2h^2}{4H^2} + \frac{9kA^2h^2}{8H^2} \\
& - \frac{3hA^3}{2H^2} + \frac{3hkA^3}{4H^2} - 6h^2 + \frac{3kh^2}{4} - 12Ah + \frac{9Ahk}{2} \\
& - 3A^2 + \frac{9kA^2}{4} - \frac{3A^2k^2}{16} - 10H^2 + 6kH^2 - \frac{3H^2k^2}{8} \\
& = -\kappa\rho
\end{aligned}
$$

$$
\begin{aligned}
x_1: \quad & \Big[-\frac{h^4}{8H^4} - \frac{Ah^3}{2H^4} - \frac{3h^2}{2H^2} + \frac{kh^2}{4H^2} - \frac{Ah}{H^2} + \frac{Ahk}{2H^2} \\
& \quad - 2 + \frac{3k}{2} - \frac{k^2}{8} \Big] \dot{A} \\
& + \Big[-\frac{Ah^3}{2H^4} - \frac{3A^2h^2}{4H^4} - \frac{3h^2}{2H^2} - \frac{3Ah}{H^2} + \frac{Ahk}{2H^2} - \frac{A^2}{2H^2} \\
& \quad + \frac{kA^2}{4H^2} - 4 + \frac{3k}{2} \Big] \dot{h} \\
& - \frac{Ah^5}{2H^4} - \frac{19A^2h^4}{16H^4} - \frac{A^3h^3}{2H^4} - \frac{15h^4}{8H^2} - \frac{6Ah^3}{H^2} + \frac{Akh^3}{2H^2} \\
& - \frac{17A^2h^2}{4H^2} + \frac{7kA^2h^2}{8H^2} - \frac{hA^3}{H^2} + \frac{hkA^3}{2H^2} - 10h^2 + \frac{9kh^2}{4} \\
& - 8Ah + 3Ahk - 3A^2 + \frac{9kA^2}{4} - \frac{3A^2k^2}{16} - 10H^2 \\
& + 6kH^2 - \frac{3H^2k^2}{8} \\
& = \kappa p_3
\end{aligned}
$$

$$
\begin{aligned}
x_0: \quad & \Big[-\frac{3Ah^3}{8H^4} - \frac{9A^2h^2}{16H^4} - \frac{9h^2}{8H^2} - \frac{9Ah}{4H^2} + \frac{3Ahk}{8H^2} - \frac{3A^2}{8H^2} \\
& \quad + \frac{3kA^2}{16H^2} - 3 + \frac{9k}{8} \Big] \dot{A} \\
& + \Big[-\frac{9A^2h^2}{16H^4} - \frac{3hA^3}{8H^4} - \frac{3h^2}{8H^2} - \frac{9Ah}{4H^2} - \frac{9A^2}{8H^2} + \frac{3kA^2}{16H^2} \\
& \quad - 3 + \frac{3k}{8} \Big] \dot{h} \\
& - \frac{9A^2h^4}{16H^4} - \frac{17A^3h^3}{16H^4} - \frac{9A^4h^2}{16H^4} - \frac{3h^4}{8H^2} - \frac{27Ah^3}{8H^2} - \frac{45A^2h^2}{8H^2} \\
& + \frac{3kA^2h^2}{16H^2} - \frac{27hA^3}{8H^2} + \frac{9hkA^3}{16H^2} - \frac{3A^4}{8H^2} + \frac{3kA^4}{16H^2} - 6h^2 \\
& + \frac{3kh^2}{8} - 9Ah + \frac{9Ahk}{8} - 6A^2 + \frac{9kA^2}{4} - 10H^2 \\
& + 3kH^2 \\
& = \kappa\bar{p}
\end{aligned}
$$

Here $\bar{p} = (p_0 + 3p_t)/4$. Because $E^{x_0}{}_{x_0} = E^{x_5}{}_{x_5}$ identically, the order-0 metric admits only sources with $p_0 = p_t$ (and no off-diagonal components); for a source with $p_0 \neq p_t$ the variation of the action with respect to the common block scale f gives the equation above with $\bar{p}$, and $p_0 - p_t$ drives a block-shape mode outside the order-0 metric (OPEN). The source is conserved,

$$\dot{\rho} = -3A(\rho + p_3) - h(\rho + p_0) - 3h(\rho + p_t) = -3A(\rho + p_3) - 4h(\rho + \bar{p}).$$

The total source is the stabiliser ($\kappa\rho_b = \kappa\rho_b^*(k/k_*)^4$, $p_3 = -\rho_b$, $\bar{p} = \rho_b$) plus matter.

5.4 The static block and the stabiliser

For $h=0$, $\dot A=0$:

$$x_4:\ -3G(k)A^2+6H^2k(1-2w_2k)+\Lambda+\kappa\rho=0,\qquad x_1:\ \text{the same with } -\kappa p_3 \text{ for } \kappa\rho,$$

$$x_0:\ 6(12w_2k-1)A^2-\frac{12(w_2-18w_3k)}{H^2}A^4+3H^2k+\Lambda-\kappa\bar p=0,\qquad G(k)=1-24w_2k+72w_3k^2=\frac{k^2-12k+16}{16}.$$

Without sources and with $A=0$ the only static block is $k_0=1/(4w_2)=8$, $\Lambda_0=-3H^2/(4w_2)=-24H^2$, with $G(k_0)=-1$ (repulsive four-dimensional gravity). With the stabiliser and $A=0$:

$$\kappa\rho_b=10H^2-6H^2k+\tfrac{3}{8}H^2k^2=\kappa\bar p=3H^2k-10H^2\implies k_\pm=12\pm\tfrac{4}{3}\sqrt{51},$$

$$k_*=12+\tfrac{4}{3}\sqrt{51}=21.5219,\quad f_*=0.215556,\quad G(k_*)=\tfrac{20}{3}+\sqrt{51}=13.8081,\quad \kappa\rho_b^*=(26+4\sqrt{51})H^2.$$

5.5 The dynamical system, the radion and the response to matter

The x_1 and x_0 components are solved for $\dot A$ and $\dot h$; with $\dot k=-2kh$ and the constraint they form a closed system for (A,h,k). Linearised about $(0,0,k_*)$ (DERIVED):

$$\delta\dot A=J_{Ak}\delta k,\quad \delta\dot h=J_{hk}\delta k,\quad \delta\dot k=-2k_*\delta h,\qquad \omega_r^2=2k_*J_{hk}=\Big(\frac{31552}{2249}-\frac{2416\sqrt{51}}{6747}\Big)H^2=(3.38705H)^2,$$

eigenvalues $0,\pm i\omega_r$: the block is stable and oscillates (the radion), the zero mode being the direction δA (a small four-dimensional cosmological term). For the stabiliser family $\bar p_b=\nu\rho_b$, $\rho_b\propto k^{2(1+\nu)}$, the static block is stable for every $\nu\in\{1/4,1/2,3/4,1,5/4,3/2,2,3\}$ ($\omega_r=3.33H$ to $4.10H$). With matter sources to first order (A^2 counts as first order: the constraint gives $3G(k_*)A^2=\kappa\delta\rho$ because $\frac{d}{dk}[6H^2k(1-2w_2k)+\kappa\rho_b]$ vanishes at k_*):

$$\delta\dot A=J_{Ak}\,\delta k+s_{A3}\,\kappa\delta p_3+s_{Ap}\,\kappa\delta\bar p+q_A\,\kappa\delta\rho,$$

$$\delta\dot h=J_{hk}\,\delta k+s_{h3}\,\kappa\delta p_3+s_{hp}\,\kappa\delta\bar p+q_h\,\kappa\delta\rho,$$

$$\delta\ddot k+\omega_r^2\,\delta k=-2k_*\big(s_{h3}\,\kappa\delta p_3+s_{hp}\,\kappa\delta\bar p+q_h\,\kappa\delta\rho\big),$$

$$J_{Ak}=\tfrac{6(238-19\sqrt{51})}{2249}H^2,\quad J_{hk}=\tfrac{4063-485\sqrt{51}}{2249}H^2,\quad s_{A3}=\tfrac{2283}{4498}-\tfrac{162\sqrt{51}}{2249},\quad s_{Ap}=\tfrac{28\sqrt{51}-114}{2249},$$

$$q_A=\tfrac{135021}{265382}-\tfrac{10308\sqrt{51}}{132691},\quad s_{h3}=\tfrac{21\sqrt{51}}{2249}-\tfrac{171}{4498},\quad s_{hp}=\tfrac{239\sqrt{51}}{6747}-\tfrac{485}{2249},\quad q_h=\tfrac{57}{4498}-\tfrac{7\sqrt{51}}{2249}$$

(numerically $0.272956H^2$, $0.266522H^2$, -0.006853, 0.038221, -0.045997, 0.028666, 0.037320, -0.009555). In the adiabatic response ($\delta\ddot k=0$), $\delta k=(-0.107556\,\kappa\delta p_3-0.140028\,\kappa\delta\bar p+0.035852\,\kappa\delta\rho)/H^2$, and 3-space obeys exactly

$$\delta\dot A=-\frac{\kappa(\delta\rho+\delta p_3)}{2G(k_*)},$$

the four-dimensional Friedmann acceleration equation (DERIVED).

5.6 The four-dimensional effective gravity

On a product of a four-dimensional metric with the static block (constant curvature $K=-H^2k$) the Lovelock densities factorise, $\mathcal{L}_{(p)}=\sum_{r+s=p}\frac{p!}{r!\,s!}\mathcal{L}^{(4)}_{(r)}\mathcal{L}^{(int)}_{(s)}$ ($p=1,2,3$ the Lovelock order), with $\mathcal{L}^{(int)}_{(s)}=(1,12K,24K^2,0)$ for $s=0,1,2,3$, $\mathcal{L}^{(4)}_{(3)}\equiv0$ and $\mathcal{L}^{(4)}_{(2)}$ topological in four dimensions, so

$$\sum_p\alpha_p\mathcal{L}_{(p)}-2\Lambda-2\kappa\rho_b=G(k)\,R^{(4)}+(\text{topological})-2G(k)\Lambda_4,\qquad \Lambda_4=\frac{6H^2k(1-2w_2k)+\Lambda+\kappa\rho_b}{G(k)},$$

and the four-dimensional field equations are those of general relativity,

$$G^{(4)}_{\mu\nu}+\Lambda_4\,g^{(4)}_{\mu\nu}=8\pi G_N\,T^{(4)}_{\mu\nu},\qquad 8\pi G_N=\frac{\kappa}{G(k_*)V_\perp},$$

$$V_\perp=\frac{2f^4v_t(1-e^{-3\eta})}{3H},\qquad T^{(4)}_{\mu\nu}=\int dx_0\,dx_5\,dx_6\,dx_7\sqrt{|g_\perp|}\,T_{\mu\nu},$$

with $\Lambda_4=0$ at the static point (the observed dark energy is a tuned residual, $\Lambda_4/H^2\sim10^{-92}$, OPEN). The gauge couplings are $g_i^2=\hat g_i^2/V_\perp$.

6 The field equations in the hidden coordinate: the masses

On the static block every four-dimensional mass is an eigenvalue of the x_0 operator of its field (the author's requirement: the wave functions depend on x_0 periodically with period $\pi/(6H)$, through $y(z)$). With $\xi = y/L \in [-1, 0]$ on each half of a period, bosonic eigenvalues are $\Lambda = (m\,fL)^2$ and fermionic ones $e = \varepsilon\, fL$.

6.1 Gauge fields and the graviton

A four-dimensional component $A_\mu(x)u(y)$ (gauge $A_y = 0$) has the reduced action $\int_{-L}^{0} dy\, f^4 e^{3Hy}\big[(\partial_y u)^2/f^2 - m^2u^2\big]$ per half (times $1/\hat g^2$), hence (DERIVED)

$$-\frac{1}{f^2}e^{-3Hy}\partial_y\big(e^{3Hy}\partial_y u\big) = m^2 u \iff -e^{-3\eta\xi}\big(e^{3\eta\xi}u'\big)' = \Lambda u,$$

$$\text{fold: } u'(0) = 0 \text{ (even) or } u(0) = 0 \text{ (odd)}; \qquad \text{tip: } u'(-1) = \beta\, u(-1),$$

the tip condition being the natural boundary condition of the tip mass term: $\beta = 0$ for the photon and the gluons, β_W for W^+ and W^- (one value for both: they are particle and antiparticle), $\beta_Z = \beta_W/\cos^2\theta_W$ for the Z (the ratio of the coefficients $\hat g^2$ and $\hat g^2 + \hat g'^2$ of the tip mass terms; θ_W is the angle of the couplings, $\cos^2\theta_W = \hat g^2/(\hat g^2 + \hat g'^2)$, fixed in UM-2 by the measured masses and the measured electroweak parameter ρ_0 as $\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0$). The transverse-traceless graviton obeys the same operator with $\beta = 0$ in Einstein gravity. With $u = e^{-3\eta\xi/2}\tilde u$, $q^2 = \Lambda - 9\eta^2/4$:

$$\text{even: } \Big[q^2 + \frac{3\eta}{2}\Big(\beta + \frac{3\eta}{2}\Big)\Big]\sin q - \beta q\cos q = 0, \qquad \text{odd: } q\cos q + \Big(\beta + \frac{3\eta}{2}\Big)\sin q = 0.$$

For $\beta = 0$: $\Lambda = 0$ (the flat profile; graviton, photon and gluons are exactly massless), $9\eta^2/4 + n^2\pi^2$ (even), $q\cot q = -3\eta/2$ (odd).

6.2 Fermions

With $\Psi = e^{ikx_1}e^{-\frac{3}{2}Hy}e^{a_4/2}b_4^{-1}V_\beta\,\chi(y, x_4)$ the eight-dimensional Dirac equation (canonical spin connection included) is EXACTLY (DERIVED, all blocks, general a_4, b_4, $M(y)$, $v(y)$)

$$\gamma^M D_M\Psi - \big(M(y) - iv(y)\gamma^{x_4}\big)\Psi = e^{ikx_1}e^{-\frac{3}{2}Hy}e^{a_4/2}b_4^{-1}\,\gamma^{x_4}V_\beta\big[\partial_4\chi + ih_j\chi\big],$$

$$h_j = j\Big[-\frac{i}{f}\sigma_1\partial_y + M(y)\sigma_2 + e^{-a_4}k\,\sigma_3\Big] + v(y),$$

so $i\partial_4\chi = h_j\chi$. In every block (j, s_2, s_3): $\gamma^{x_0} = \sigma_3$, $B = js_2$, $C = s_2\sigma_2$, $\Gamma_4 = -js_2\sigma_3$, $P_+ = (1 - B\sigma_3)/2$. For a Standard-Model multiplet $M = \mu_\psi$, $v = 0$; with $c = f\mu_\psi L$ the zero modes are $\chi = (e^{c\xi}, 0)$ (doublet, $\Gamma_4 = +1$) and $\chi = (0, e^{-c\xi})$ (singlet, $\Gamma_4 = -1$), with $h_j(e^{c\xi}, 0) = je^{-a_4}k\,(e^{c\xi}, 0)$ exactly (massless four-dimensional particles). At $k = 0$ in the block $j = +1$ the components of $\chi = (a, ib)$ obey $a' = c\,a - e\,b$, $b' = e\,a - c\,b$ ($e = \varepsilon fL$); the doublet ($c_L > 0$) and singlet ($c_R < 0$) solutions that satisfy the fold conditions reach the tip with

$$(a_L, b_L) = (C_L - c_L s_L,\ -s_L e), \qquad (a_R, b_R) = (s_R e,\ C_R + c_R s_R), \qquad s = \frac{\sinh w}{w}, \quad C = \cosh w, \quad w = \sqrt{c^2 - e^2},$$

and the tip mixing conditions of one generation give the exact mass condition

$$D(e) = e^2 s_L s_R - \kappa_f^2\,(C_L - c_L s_L)(C_R + c_R s_R) = 0,$$

$$e \simeq \kappa_f\,\zeta(c_L)\,\zeta(-c_R), \qquad \zeta(c)^2 = \frac{c\,e^{-c}}{\sinh c} \qquad (c_L > 0,\ c_R < 0).$$

For three generations, with doublet amplitudes $\mathcal{A}_\alpha$ and singlet amplitudes $\mathcal{B}_i$, the matrix conditions $\mathrm{diag}(b_{L\alpha})\mathcal{A} = \kappa\,\mathrm{diag}(b_{Ri})\mathcal{B}$, $\mathrm{diag}(a_{Ri})\mathcal{B} = -\kappa^\dagger\mathrm{diag}(a_{L\alpha})\mathcal{A}$ give

$$\Big[\mathrm{diag}(b_{L\alpha}) + \kappa\,\mathrm{diag}\Big(\frac{b_{Ri}}{a_{Ri}}\Big)\kappa^\dagger\,\mathrm{diag}(a_{L\alpha})\Big]\mathcal{A} = 0,$$

the masses being the roots of its determinant and $\mathcal{A}$ the left-handed flavour content. For a common doublet bulk mass and a unitary κ (DERIVED)

$$\det\Big[\dots\Big]=\prod_{i=1}^{3}\frac{b_La_{Ri}+a_Lb_{Ri}}{a_{Ri}},\qquad b_La_R+a_Lb_R=-D(e)\big|_{\kappa_f=1},\qquad \mathcal{A}^{(i)}=\kappa\,\hat{e}_i,\quad \mathcal{B}^{(i)}\propto\hat{e}_i:$$

the masses are the one-generation roots with $\kappa_f=1$, the left-handed rotation of each sector is its tip matrix (so that $U=\kappa_e^\dagger\kappa_\nu$ and $V=\kappa_u^\dagger\kappa_d$), and the right-handed rotations are trivial (verified with mpmath at 50 digits to 3.0×10^{-42}, the two sets of roots refined in the same computation; the masses stored by the mass computation agree with them to 4.5×10^{-11}).

7 dirac16complex: the Kohn-Sham equations coupled to all fields

The interacting many-body state of Ψ on the block is described by Kohn-Sham orbitals of

$$h_j\chi=\varepsilon\chi,\qquad M_{eff}(y)=m+\frac{15}{16}\lambda S(y)+\frac{\langle\mathcal{F}\rangle}{\Lambda_\star^3},\qquad v(y)=-\frac{\lambda}{16}n(y)+\frac{c_V}{\Lambda_\star^6}\sum_{i=1}^{3}\sum_{\psi}q_\psi n_{\psi i}(y),$$

where $\langle\mathcal{F}\rangle=c_G\langle G^\alpha_{MN}G^{\alpha MN}\rangle+\dots$ is the gauge-field condensate (in the QCD vacuum the gluon condensate, a uniform mass shift), $n_{\psi i}$ the number densities of the Standard-Model fermions of the three generations ($\langle\bar\psi_i\gamma^{x_4}\psi_i\rangle=i\,n_{\psi i}$; I11). The densities are

$$n=\sum_o w\,g\,f_o\,n_o,\quad S=\sum_o w\,g\,f_o\,s_o,\qquad n_o=P\chi_o^\dagger\chi_o,\quad s_o=P\chi_o^\dagger j\sigma_2\chi_o,\quad t_o=P\chi_o^\dagger j\sigma_3\chi_o,\quad P=\frac{e^{-3Hy}}{\mathrm{Vol}_7},$$

with $w=1/2$ for each fold parity, $g=4r_3(|\vec n|^2)$ the degeneracy of the momentum shell $\vec k=\Delta k\,\vec n$ ($\vec n\in\mathbb{Z}^3$, $r_3(|\vec n|^2)$ the number of representations of $|\vec n|^2$ as a sum of three squares), $\mathrm{Vol}_7=(2\pi/\Delta k)^3v_tf^4$ (at $a_4=0$; v_t the coordinate volume of the extra times) the normalisation volume of a cell, of dimension length^6 (the proper seven-volume of a cell, with its hidden extent, is $2\,\mathrm{Vol}_7(1-e^{-3\eta})/(3H)$), and occupations f_o (aufbau, or Mermin $f_o=1/(e^{(\varepsilon_o-\mu)/T}+1)$). The energy and the energy-momentum tensor are

$$E_{KS}=\sum_o g\,f_o\,\varepsilon_o-2\,\mathrm{Vol}_7\int_{-L}^{0}e^{3Hy}e_{int}\,dy,\qquad e_{int}=\frac{15}{32}\lambda S^2-\frac{1}{32}\lambda n^2,$$

$$\rho=\sum_o wgf_o\varepsilon_o\,n_o-e_{int},\qquad p_3=\sum_o wgf_o\,\frac{e^{-a_4|k|}}{3}t_o+e_{int},\qquad p_t=e_{int},$$

$$p_0=\sum_o wgf_o\big[(\varepsilon_o-v)n_o-M_{eff}s_o-e^{-a_4|k|}t_o\big]+e_{int},$$

which satisfy $E_{KS}=2\mathrm{Vol}_7\int e^{3Hy}\rho\,dy$ and the y-conservation law $\big[e^{3Hy}p_0\big]_{-L}^{0}=3H\int_{-L}^{0}e^{3Hy}p_t\,dy$ (DERIVED for every self-consistent state). Boundary conditions: fold $\chi_2(0)=0$ (even) or $\chi_1(0)=0$ (odd), tip $\chi_2(-L)=0$ (from $(1\mp\gamma^{x_0})\Psi=0$). The Kohn-Sham source enters the background equations through its block averages ρ, p_3 and $\bar p=(p_0+3p_t)/4$.

7.1 The portals on the zero modes and the mean fields of an environment (version 3)

In the block form $\gamma^{x_0}=\sigma_3$, so I13 adds to the Kohn-Sham Hamiltonian a potential on the lower component only:

$$h_j\;\to\;h_j+P_b\,v_b(y),\qquad P_b=\tfrac12(1-\sigma_3),\qquad v_b(y)=\frac{c_A}{\Lambda_\star^6}\,\frac13\sum_q n_q(y),$$

with $n_q(y)$ the proper eight-dimensional density of the quarks. On the zero modes the portals become four-dimensional contact interactions. A four-dimensional particle of density n (per 3-volume) in an orbital or zero mode with the normalised profile $\varrho(y)$ ($\int_{-L}^{0}\varrho\,dy=1$) has the proper eight-dimensional density $n\,\varrho(y)\,e^{-3Hy}/(2f^4v_t)$ on

each half of the period (proper transverse volume element $f^4e^{3Hy}dy\,d^3x_t$, two halves; v_t the coordinate volume of the extra times), so a state o of Ψ (orbital (a_o,b_o), $\int_{-L}^0(a_o^2+b_o^2)\,dy=1$) and a chirality component f of a Standard-Model fermion (zero-mode profile ϱ_f; $\varrho_c(y)=(2c/L)e^{2cy/L}/(1-e^{-2c})$ for the localisation c) interact with

$$\mathcal{L}_4=\big(G_V q_f\tilde I_{of}+G_A q_f^B\tilde I_{of}^{(b)}\big)\,(i\bar\chi_o\gamma^\mu\chi_o)(i\bar f\gamma_\mu f),\qquad \tilde I_{of}=\int_{-L}^0(a_o^2+b_o^2)\varrho_f e^{-3Hy}dy,\qquad \tilde I_{of}^{(b)}=\int_{-L}^0 b_o^2\varrho_f e^{-3Hy}dy,$$

$$G_V=\frac{c_V H}{2f^4v_t\Lambda_\star^6},\qquad G_A=\frac{c_A H}{2f^4v_t\Lambda_\star^6},\qquad q_f^B=\tfrac13\ \text{(quarks)},\ 0\ \text{(leptons)}$$

($\tilde I$ in units of H). $\tilde I_{of}$ and $\tilde I_{of}^{(b)}$ are exactly the responses $d\varepsilon_o/de$ of the Kohn-Sham level to the mean fields $v=e\,\varrho_f(y)e^{-3Hy}$ and $v_b=e\,\varrho_f(y)e^{-3Hy}$ (Hellmann-Feynman). I12 becomes $G_i\bar\chi\chi F^i_{\mu\nu}F^{i\mu\nu}$ (canonical fields) with $G_i=\mathcal{O}\,c_i g_i^2/\Lambda_\star^3$, $\mathcal{O}=\partial\varepsilon_{DM}/\partial m$.

In an environment with the net densities n_p, n_n, n_e of protons, neutrons and electrons (unpolarised: each chirality carries half; $n_u=2n_p+n_n$, $n_d=n_p+2n_n$) and the temperature T, the level ε_o moves by

$$\delta\varepsilon_o=\sum_f\big(G_V q_f\tilde I_{of}+G_A q_f^B\tilde I_{of}^{(b)}\big)\,n_f+\mathcal{O}\,\frac{c}{\Lambda_\star^3}\Big[\langle g_s^2G^\alpha G^\alpha\rangle+\langle g^2W^aW^a\rangle+\langle g'^2BB\rangle\Big]$$

(universal $c_G=c_W=c_B=c$; canonical field strengths), with the gluon condensate from the trace anomaly of QCD, $\langle\theta^\mu{}_\mu\rangle=\rho-3p=-\frac{b\alpha_s}{8\pi}\langle G^\alpha_{\mu\nu}G^{\alpha\mu\nu}\rangle$ for the thermal parts (written in the signature $(+,-,-,-)$, in which the trace of the energy-momentum tensor of a fluid is $\rho-3p$; in the signature $(+,+,+,-)$ of this document the trace is $3p-\rho=+\frac{b\alpha_s}{8\pi}G^\alpha_{\mu\nu}G^{\alpha\mu\nu}$, while $G^\alpha_{\mu\nu}G^{\alpha\mu\nu}$ is the same in both, so the relation below does not depend on the convention),

$$\Big\langle\frac{\alpha_s}{\pi}G^\alpha G^\alpha\Big\rangle_T=\Big\langle\frac{\alpha_s}{\pi}G^\alpha G^\alpha\Big\rangle_0-\frac{8}{b}(\rho-3p)_{QCD},\qquad b=11-\frac23 n_f,\qquad \langle g_s^2G^2\rangle=4\pi^2\Big\langle\frac{\alpha_s}{\pi}G^2\Big\rangle,$$

$\langle\frac{\alpha_s}{\pi}G^2\rangle_0=0.012$ GeV^4 (STANDARD), the QCD interaction measure $(\rho-3p)/T^4=4\exp[-\ln^2(T/0.2\text{ GeV})/0.5]$ below 1 GeV (ESTIMATE) and $\frac{b}{3}(1+\frac{5n_f}{12})\alpha_s^2$ above (α_s at the scale $2\pi T$, one loop from $\alpha_s(m_Z)$), which follows from the pressure $p=\frac{8\pi^2}{45}T^4[1+\frac{21n_f}{32}-\frac{15}{4}(1+\frac{5n_f}{12})\frac{\alpha_s}{\pi}]$ and $d\alpha_s/d\ln T=-\frac{b}{2\pi}\alpha_s^2$ through $\rho-3p=T^5\,d(p/T^4)/dT$; and for each massive vector boson V of mass m_V the thermal condensate

$$\langle F_{\mu\nu}F^{\mu\nu}\rangle_T=-2(\rho-3p)_V,\qquad (\rho-3p)_V=\rho_V-3p_V\ \text{of its ideal gas (3 states)},$$

because on shell the Proca Lagrangian, written in the signature $(+,-,-,-)$ as $-\frac14F^2+\frac12m_V^2A_\mu A^\mu$, is a total derivative (so $\langle-\frac14F^2\rangle=-\langle\frac12m_V^2A_\mu A^\mu\rangle$), while $T^\mu{}_\mu=-m_V^2A_\mu A^\mu=\rho-3p$ (in the signature $(+,+,+,-)$ the mass term is $-\frac12m_V^2A_\mu A^\mu$, $\langle-\frac14F^2\rangle=+\langle\frac12m_V^2A_\mu A^\mu\rangle$ and $T^\mu{}_\mu=-m_V^2A_\mu A^\mu=3p-\rho$: the relation is the same). This is the free part. Above m_W the interaction pressure of the electroweak plasma dominates: for a gauge field with the Lagrangian $-F_{geo}^2/(4g^2)$ ($F_{geo}=gF$) the Feynman-Hellmann theorem gives

$$\langle F_{geo}^2\rangle=-4\,\frac{\partial p}{\partial(1/g^2)}=4g^2p_2,\qquad p_2=-\frac{g^2T^4}{144}\,d_A\Big[C_A+\frac52T_FN_f\Big]$$

for the leading-order pressure $p_2\propto g^2$ of a gauge plasma with N_f Dirac fermions in the representation of index T_F, massless (STANDARD; for QCD, $d_A=8$, $C_A=3$, $T_F=\frac12$, $N_f=n_f$, it reproduces the trace-anomaly condensate above exactly, DERIVED to 10^{-16}): for $SU(2)$ $d_A=3$, $C_A=2$ and twelve Weyl doublets ($T_FN_f=3$); for $U(1)_Y$ $d_A=1$, $C_A=0$ and $\frac52T_FN_f\to\frac52\sum_{Dirac}Y^2=\frac52\cdot5$ ($\sum_{Weyl}Y^2=10$ for three generations); no scalar term (ESTIMATE), and below m_W multiplied by the thermal suppression $\rho(m_W,T)/\rho(0,T)$ of a massive vector (ESTIMATE). The couplings are taken at $\mu=2\pi T$, one loop from the $\overline{MS}$ values at m_Z without a Higgs particle,

$$\frac{1}{\alpha_i(\mu)}=\frac{1}{\alpha_i(m_Z)}+\frac{b_i}{2\pi}\ln\frac{\mu}{m_Z},\qquad b_2=\frac{10}{3},\quad b_Y=-\frac{20}{3},\qquad \frac{1}{\alpha_2(m_Z)}=\frac{\hat s_Z^2}{\alpha},\quad \frac{1}{\alpha_Y(m_Z)}=\frac{1-\hat s_Z^2}{\alpha}$$

($\alpha=1/127.95$ at m_Z and the $\overline{MS}$ value $\hat s_Z^2=\sin^2\hat\theta_W(m_Z)=0.23122$; for QCD $b=11-\frac23n_f$). Hence

$$\langle g^2W^aW^a\rangle=g^2\big[2(-2I_W)+\cos^2\theta_W(-2I_Z)\big]+4g^2p_2^{SU(2)},$$

$$\langle g'^2 BB\rangle = g'^2\sin^2\theta_W(-2I_Z) + 4g'^2 p_2^{U(1)}, \qquad I_V = (\rho-3p)_V$$

($g^2 = 4\pi\alpha_2(2\pi T)$, $g'^2 = 4\pi\alpha_Y(2\pi T)$; here θ_W is the angle of the couplings of section 6.1). Conversely Ψ acts on the Standard Model through the field-dependent couplings of section 3.1, $\delta\alpha_i/\alpha_i = -4G_i n_{DM}$, and on the x_0 problem of the W and Z through the kinetic function $1 + e\,s_{DM}(y)$,

$$-\frac{1}{Ke^{3\eta\xi}}\big(Ke^{3\eta\xi}u'\big)' = \Lambda u, \qquad K(\xi) = 1 + e\,s_{DM}(y), \qquad u'(0) = 0, \qquad K(-1)u'(-1) = \beta u(-1),$$

$s_{DM} = 2ab$ the scalar density of the dark-matter orbital (primes $d/d\xi$).

Gravity closes the loop. The block responds to the block-averaged sources ρ, p_3, $\bar p$ of all species with the adiabatic response of section 5.5 ($\delta\ddot k = 0$),

$$\delta k = \frac{c_\rho\,\kappa\delta\rho + c_3\,\kappa\delta p_3 + c_p\,\kappa\delta\bar p}{H^2}, \qquad c_\rho = 0.035852, \quad c_3 = -0.107556, \quad c_p = -0.140028,$$

$$\delta\ln f = -\frac{\delta k}{2k_*}, \qquad \delta\ln m_s = \frac{d\ln m_s}{d\ln f}\,\delta\ln f,$$

iterated with the masses until self-consistency. Radiation ($p_3 = \rho/3$) drops out exactly ($c_3 = -3c_\rho$), so only the trace $\rho - 3p_3$ and the hidden pressure act; the block averages of four-dimensional densities are $\kappa\delta\rho = 8\pi G_N G(k_*)\rho_4$ (likewise for p_3 and $\bar p$; section 5.6). The hidden pressure is the response of the pressure to the block scale at fixed temperature, $\bar p = \frac14\,\partial p/\partial\ln f$:

$$\bar p = -\frac14\sum_s(\rho_s - 3p_s)\frac{d\ln m_s}{d\ln f} - \frac14(\rho-3p)_{QCD}\frac{d\ln\Lambda^{(n_f)}}{d\ln f} + \frac14\sum_{i=2,Y} p_2^{(i)}\,\frac{d\ln\alpha_i}{d\ln f}.$$

The first sum runs over the species of mass $m_s(f)$ (for an ideal gas $m\,\partial p/\partial m = -(\rho-3p)$; non-relativistic matter gives $-\frac14\rho\,d\ln m/d\ln f$): the plasma species as ideal gases with their x_0 masses, the net nucleons and electrons (degenerate Fermi gases in the cold environments, $\rho = nm$ and $p = nT$ in the hot ones) and the dark matter (in equilibrium with eight states while $M/T < 40$, its relic density afterwards). The second term follows from $p = T^4\varphi(T/\Lambda)$, for which $\partial p/\partial\ln\Lambda = -(\rho-3p)$; the third from the order-α pressure $p_2^{(i)} \propto \alpha_i$ of the electroweak plasma, with

$$\frac{d\ln\alpha_i(\mu)}{d\ln f} = -\alpha_i(\mu)\Big[\frac{4}{\alpha_i(m_{KK})} + \frac{b_i}{2\pi}\Big]$$

(the bulk gauge couplings scale as $\alpha_i(m_{KK}) \propto f^{-4}$, since $g_i^2 = \hat g_i^2/V_\perp$ and $V_\perp \propto f^4$, and $m_{KK} \propto 1/f$). In the trace $\rho - 3p_3$ the QCD and the electroweak plasmas contribute $(\rho-3p)_{QCD}$ and $(\rho-3p)_i = -\frac{b_i\alpha_i}{2\pi}\,p_2^{(i)}$ (the leading-order trace anomaly, from $\rho - 3p = T^5\,d(p/T^4)/dT$ with $d\ln\alpha_i/d\ln\mu = -b_i\alpha_i/(2\pi)$).

The environments of the coupled solution: the cosmic mean today (the Planck baryon density with the helium fraction $Y_p = 0.2457$ and $\Omega_c h^2 = 0.120$), the Galaxy at the Sun ($n_p = n_e = 1$ cm^{-3}), the Earth (mean density 5.51 g/cm^3, 48 % protons), the centre of the Sun (150 g/cm^3, 66 % protons, 32 % neutrons), a white dwarf (10^6 g/cm^3 of carbon and oxygen, equal numbers of protons and neutrons), a neutron-star core (0.3 fm^{-3}, 10 % protons, $n_e = n_p$), all bound objects with the local dark-matter density 0.4 GeV/cm^3; and the hot Universe at nucleosynthesis ($T = 1$ MeV), the QCD crossover (200 MeV), the electroweak era (100 GeV) and the dark-matter freeze-out ($T = M/x_f$), with the net baryon density $\eta_b n_\gamma$, $\eta_b = 6.1\times10^{-10}$, $n_\gamma = 2\zeta(3)T^3/\pi^2$, shared as $n_n/n_p = e^{-Q/T}$ ($Q = 1.293$ MeV), $n_e = n_p$, and the dark matter in equilibrium, $n = 2\cdot4(MT/2\pi)^{3/2}e^{-M/T}$ for the two species together, while $M/T < 40$.

The nucleon follows the QCD scale: with $1/\alpha_s(m_{KK}) = 1/\alpha_s(m_Z) + \frac{7}{2\pi}\ln(m_{KK}/m_Z)$ (six flavours, one loop), $\alpha_s \propto f^{-4}$, $m_{KK} \propto 1/f$, $\Lambda^{(6)} = m_{KK}e^{-2\pi/(7\alpha_s(m_{KK}))}$, the one-loop matching $\Lambda^{(3)} = \Lambda^{(6)\,21/27}(m_c m_b m_t)^{2/27}$ and $m_p = (1 - \frac{\sigma_{\pi N}+\sigma_s}{m_p})\,C\Lambda^{(3)} + \sigma$ terms,

$$\frac{d\ln m_p}{d\ln f} = \Big(1 - \frac{\sigma_{\pi N}+\sigma_s}{m_p}\Big)\Big[\frac{21}{27}\Big(-1 - \frac{8\pi}{7\alpha_s(m_{KK})}\Big) + \frac{2}{27}\sum_{q=c,b,t}\frac{d\ln m_q}{d\ln f}\Big] + \frac{\sigma_{\pi N}}{2m_p}\Big(\frac{d\ln m_u}{d\ln f} + \frac{d\ln m_d}{d\ln f}\Big) + \frac{\sigma_s}{m_p}\frac{d\ln m_s}{d\ln f}$$

($\sigma_{\pi N} = 45$ MeV, $\sigma_s = 40$ MeV; ESTIMATE). Through the thresholds,

$$\frac{d\ln\Lambda^{(6)}}{d\ln f} = -1 - \frac{8\pi}{7\alpha_s(m_{KK})}, \qquad \frac{d\ln\Lambda^{(5)}}{d\ln f} = \frac{21}{23}\frac{d\ln\Lambda^{(6)}}{d\ln f} + \frac{2}{23}\frac{d\ln m_t}{d\ln f},$$

$$\frac{d\ln\Lambda^{(4)}}{d\ln f}=\frac{21}{25}\frac{d\ln\Lambda^{(6)}}{d\ln f}+\frac{2}{25}\Big(\frac{d\ln m_t}{d\ln f}+\frac{d\ln m_b}{d\ln f}\Big),\qquad \frac{d\ln\Lambda^{(3)}}{d\ln f}=\frac{21}{27}\frac{d\ln\Lambda^{(6)}}{d\ln f}+\frac{2}{27}\sum_{q=c,b,t}\frac{d\ln m_q}{d\ln f},$$

$$\frac{d\ln m}{d\ln f}=-1+\frac{1}{2}\Big[\frac{d\ln\zeta(c_L)^2}{d\ln c_L}+\frac{d\ln\zeta(-c_R)^2}{d\ln(-c_R)}\Big]$$

for a fermion of mass $m=e/(fL)$ with the first-order root $e\simeq\kappa_f\zeta(c_L)\zeta(-c_R)$ of section 6.2 and localisations $c=f\mu L$ proportional to f (DERIVED): $d\ln m_q/d\ln f=-18.82$, -18.01, -14.90, -12.15, -10.89, -6.77 for u, d, s, c, b, t, so that $\alpha_s(m_{KK})=0.0606$, $d\ln\Lambda^{(6)}/d\ln f=-60.2$, $d\ln\Lambda^{(3)}/d\ln f=-49.0$ and $d\ln m_p/d\ln f=-46.1$.

8 The linear perturbations (order ϵ): CMB, CNB, BAO

The x_0-independent ($n=0$) perturbations obey the four-dimensional Einstein equations of section 5.6 linearised about a flat Friedmann background; the $n\geq1$ modes (Kaluza-Klein gravitons and gauge bosons, mass ≥99 TeV) and the radion (48 TeV) are absent at cosmological scales. In conformal time τ ($dx_4=a\,d\tau$, $a=e^{a_4}$, $\mathcal{H}=a'/a$) and the Newtonian gauge $ds^2=a^2\big[-(1+2\psi)d\tau^2+(1-2\phi)d\vec{x}^2\big]$ (UM-2: $\psi=\epsilon\Phi$, $\phi=\epsilon\Psi_N$), for a Fourier mode k (STANDARD, Ma and Bertschinger):

$$k^2\phi+3\mathcal{H}(\phi'+\mathcal{H}\psi)=-4\pi G_N a^2\delta\rho,\qquad k^2(\phi'+\mathcal{H}\psi)=4\pi G_N a^2(\bar\rho+\bar P)\theta,$$

$$\phi''+\mathcal{H}(\psi'+2\phi')+(2\mathcal{H}'+\mathcal{H}^2)\psi+\frac{k^2}{3}(\phi-\psi)=4\pi G_N a^2\delta P,\qquad k^2(\phi-\psi)=12\pi G_N a^2(\bar\rho+\bar P)\sigma,$$

summed over the species of UM-2:

- **dark matter** (the odd quanta of Ψ_A and Ψ_B, cold): $\delta_c'=-\theta_c+3\phi'$, $\theta_c'=-\mathcal{H}\theta_c+k^2\psi$;
- **baryons** (u, d confined in protons and neutrons; electrons): $\delta_b'=-\theta_b+3\phi'$, $\theta_b'=-\mathcal{H}\theta_b+c_s^2k^2\delta_b+\frac{4\bar\rho_\gamma}{3\bar\rho_b}an_e\sigma_T(\theta_\gamma-\theta_b)+k^2\psi$;
- **photons**: $\delta_\gamma'=-\frac{4}{3}\theta_\gamma+4\phi'$, $\theta_\gamma'=k^2(\frac{1}{4}\delta_\gamma-\sigma_\gamma)+k^2\psi+an_e\sigma_T(\theta_b-\theta_\gamma)$, $F_{\gamma2}'=2\sigma_\gamma'=\frac{8}{15}\theta_\gamma-\frac{3}{5}kF_{\gamma3}-\frac{9}{5}an_e\sigma_T\sigma_\gamma+\frac{1}{10}an_e\sigma_T(G_{\gamma0}+G_{\gamma2})$, $F_{\gamma l}'=\frac{k}{2l+1}[lF_{\gamma(l-1)}-(l+1)F_{\gamma(l+1)}]-an_e\sigma_TF_{\gamma l}$ ($l\geq3$), $G_{\gamma l}'=\frac{k}{2l+1}[lG_{\gamma(l-1)}-(l+1)G_{\gamma(l+1)}]+an_e\sigma_T[-G_{\gamma l}+\frac{1}{2}(F_{\gamma2}+G_{\gamma0}+G_{\gamma2})(\delta_{l0}+\frac{\delta_{l2}}{5})]$;
- **the three neutrino mass eigenstates** ν_i, $i=1,2,3$ (masses $m_i=1.000$, 8.712, 50.140 meV; after decoupling the flavour coherence is lost and each mass state propagates on its own; comoving momentum q, comoving energy $E_i=\sqrt{q^2+a^2m_i^2}$, Fermi-Dirac distribution $\bar f_\nu(q)$, multipoles $\mathcal{N}_l^{(i)}$ of the perturbed distribution): $\mathcal{N}_0^{(i)\prime}=-\frac{qk}{E_i}\mathcal{N}_1^{(i)}-\phi'\frac{d\ln\bar f_\nu}{d\ln q}$, $\mathcal{N}_1^{(i)\prime}=\frac{qk}{3E_i}(\mathcal{N}_0^{(i)}-2\mathcal{N}_2^{(i)})-\frac{E_ik}{3q}\psi\frac{d\ln\bar f_\nu}{d\ln q}$, $\mathcal{N}_l^{(i)\prime}=\frac{qk}{(2l+1)E_i}[l\mathcal{N}_{l-1}^{(i)}-(l+1)\mathcal{N}_{l+1}^{(i)}]$ ($l\geq2$);
- the fold bands of Ψ are never thermalised; produced by freeze-in through I11 (section 10), they are massless free-streaming radiation with $\Delta N_{eff}=0.001$ (below), the only dark radiation of UM-2.

The background is $\mathcal{H}^2=\frac{8\pi G_N}{3}a^2\bar\rho_{tot}+\frac{\Lambda_4}{3}a^2$, adiabatic initial conditions with the power spectrum $\mathcal{P}(k)=A_s(k/k_0)^{n_s-1}$. The relativistic content is $N_{eff}=3.0432$, carried by the three mass eigenstates (each with the degeneracy $N_{eff}/3=1.0144$ at the temperature $T_\nu=(4/11)^{1/3}T_\gamma$ of instantaneous decoupling, which represents the spectrum heated by e^+e^- annihilation; the adopted $N_{eff}=3.0432$ is the precision value, close to the 3.0476 to 3.0485 computed in section 9.4); in version 3 the fold bands produced by freeze-in (section 10) add massless free-streaming radiation with $\Delta N_{eff}=0.001$, obeying the massless limit of the neutrino hierarchy. The rms potential per logarithmic wavenumber is $\Delta_\Phi(k)=[k^3P_\Phi(k)/(2\pi^2)]^{1/2}$; inside the horizon the Poisson equation $\Phi=\frac{3}{2}\Omega_mH_0^2\delta_m/(ak^2)$ holds, and on the largest scales in the matter era $\Phi=\frac{3}{5}\zeta$ (ζ the primordial curvature perturbation), suppressed by Λ_4 with the growth factor $g(a)=D(a)/a$, $D(a)=\frac{5}{2}\Omega_mE(a)\int_0^a db/[bE(b)]^3$, $E(a)=(\Omega_ma^{-3}+1-\Omega_m)^{1/2}$. The amplitude of the matter fluctuations is $\sigma_8^2=\int_0^\infty\frac{dk}{k}\frac{k^3P_m(k)}{2\pi^2}W^2(kR)$ with the top hat $W(x)=3(\sin x-x\cos x)/x^3$, $R=8h^{-1}$ Mpc.

8.1 Beyond $n=0$: the Kaluza-Klein gravitons and the radion (version 3)

The $n\geq1$ terms of $h_{MN}^{(1)}=\sum_n h_{MN}^{(1,n)}u_n(x_0)$ are massive. For the graviton ($\beta=0$) the even profiles and masses are

$$u_n(\xi)=e^{-a\xi}\Big(\cos q_n\xi+\frac{a}{q_n}\sin q_n\xi\Big),\qquad q_n=n\pi,\qquad a=\frac{3\eta}{2},\qquad m_n=\frac{\sqrt{\Lambda_n}}{fL},\quad \Lambda_n=\frac{9\eta^2}{4}+n^2\pi^2,$$

normalised by $N_n = \int_{-1}^{0} e^{3\eta\xi} u_n^2 d\xi = \frac{1}{2}(1 + a^2/q_n^2)$ ($N_0 = (1 - e^{-3\eta})/(3\eta)$ for $u_0 = 1$). Each mode couples to a source with the normalised x_0 profile ϱ ($\int_{-1}^{0} \varrho\, d\xi = 1$) through $\int \varrho\, u_n\, d\xi$, so a static source M with the profile ϱ_M attracts a mass m with the profile ϱ_m with

$$V(r) = -\frac{G_N M m}{r}\Big[1 + \sum_{n\geq 1} \alpha_n e^{-m_n r}\Big], \qquad \alpha_n = \frac{N_0}{N_n}\Big(\int_{-1}^{0} \varrho_M\, u_n\, d\xi\Big)\Big(\int_{-1}^{0} \varrho_m\, u_n\, d\xi\Big)$$

($(N_0/N_n)(\int \varrho\, u_n\, d\xi)^2$ for two masses with the same profile); a source with the flat profile $\varrho \propto e^{3\eta\xi}$ of a gauge zero mode has $\alpha_n = 0$ (orthogonality of u_n to u_0). The radion $\sigma = \delta \ln f$ obeys the linearised block equations of section 5.5 with the sources of matter (its trace $\rho - 3p_3$ and its hidden pressure); its static response is $\delta \ln f$ of section 7.1, its mass $m_r = \omega_r$.

9 The flavour evolution of the three neutrinos

9.1 The four-fermion interactions at low energy

At momentum transfers far below m_W and m_Z the exchange of $W^{\pm}$ and Z between the zero-mode fermions (sections 2 and 4) reduces to current-current interactions. Written for the four-dimensional zero-mode fields in four-component notation, with $P_L \equiv P_+ = (1 + \Gamma_4)/2$ the projector on the chirality of the doublet zero modes and $P_R = (1 - \Gamma_4)/2$, they are (STANDARD)

$$\mathcal{L}_{CC} = -2\sqrt{2}\, G_F \left(\bar{\nu}_\alpha \gamma^\mu P_L \ell_\alpha\right)\left(\bar{\ell}_\beta \gamma_\mu P_L \nu_\beta\right), \qquad \mathcal{L}_{NC} = -2\sqrt{2}\, G_F \left(\bar{\nu}_\alpha \gamma^\mu P_L \nu_\alpha\right) \sum_f \bar{f} \gamma_\mu \left(g_L^f P_L + g_R^f P_R\right) f,$$

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8 m_W^2}, \qquad g_L^f = T_3^f - Q^f \sin^2\theta_W, \qquad g_R^f = -Q^f \sin^2\theta_W,$$

summed over the flavours $\alpha, \beta \in \{e, \mu, \tau\}$ and the fermions f of the medium (the neutrino background is the term $V_{\nu\nu}$ of section 9.4), with $\nu_\alpha = \sum_i U_{\alpha i} \nu_i$ the flavour fields (UM-2 uses the measured $G_F = 1.1663788 \times 10^{-5}$ GeV^{-2}). A neutrino crossing a medium scatters forward coherently on it; averaged over an unpolarised medium at rest, the interactions add to its Hamiltonian the flavour-diagonal potentials

$$V_\alpha = \sqrt{2}\, G_F \Big[\delta_{\alpha e}\,(n_{e^-} - n_{e^+}) + \sum_f g_V^f\,(n_f - n_{\bar{f}})\Big], \qquad g_V^f = g_L^f + g_R^f = T_3^f - 2Q^f \sin^2\theta_W,$$

and in electrically neutral matter of electrons, protons and neutrons ($g_V^p = -g_V^e$, $g_V^n = -\frac{1}{2}$)

$$V = \mathrm{diag}(V_{CC}, 0, 0) + V_{NC}\, 1, \qquad V_{CC} = \sqrt{2}\, G_F n_e, \qquad V_{NC} = -\frac{G_F n_n}{\sqrt{2}}.$$

Antineutrinos have $-V$.

9.2 Vacuum and matter

The flavour amplitudes ψ_α of a neutrino of energy $E \gg m_i$ evolve along its path as

$$i\frac{d\psi}{dx_4} = \mathcal{H}\psi, \qquad \mathcal{H} = \frac{1}{2E}\, U\, \mathrm{diag}(m_1^2, m_2^2, m_3^2)\, U^\dagger + V,$$

with $U \to U^*$ and $V \to -V$ for antineutrinos (V_{NC} is a multiple of the identity and drops out). For a constant potential the evolution operator is $S(L) = e^{-i\mathcal{H}L}$ and $P(\nu_\alpha \to \nu_\beta) = |S_{\beta\alpha}|^2$; for a varying one it is the ordered product of the operators of thin layers. In vacuum

$$P(\nu_\alpha \to \nu_\beta) = \Big|\sum_i U_{\beta i} U_{\alpha i}^*\, e^{-i m_i^2 L/(2E)}\Big|^2,$$

$$P(\nu_\mu \to \nu_e) - P(\nu_e \to \nu_\mu) = -16 J \sin\Delta_{21} \sin\Delta_{31} \sin\Delta_{32}, \qquad \Delta_{ij} = \frac{\Delta m_{ij}^2 L}{4E},$$

with the Jarlskog invariant $J = \mathrm{Im}(U_{e2}U_{\mu3}U_{e3}^*U_{\mu2}^*) = c_{12}s_{12}c_{23}s_{23}c_{13}^2 s_{13}\sin\delta$ and CPT, $P(\nu_\alpha \to \nu_\beta) = \bar{P}(\bar{\nu}_\beta \to \bar{\nu}_\alpha)$. In matter the mixing is resonant (the MSW effect) where the potential cancels the vacuum splitting of a sector: for the 1-3 sector, after the rotation R_{23} (which commutes with V),

$$2EV_{CC} = \Delta m^2_{ee}\cos 2\theta_{13}, \qquad \Delta m^2_{ee} = c_{12}^2\Delta m^2_{31} + s_{12}^2\Delta m^2_{32},$$

and for the 1-2 sector, after the rotation by θ_{13}, which leaves the potential $V_{CC}c_{13}^2$,

$$2EV_{CC}c_{13}^2 = \Delta m^2_{21}\cos 2\theta_{12}.$$

A neutrino produced at a density where the effective 1-2 mixing angle is θ_{12}^m and propagating adiabatically to the vacuum arrives, averaged over the incoherent mass states, with

$$P_{ee} = c_{13}^4\Big(\frac{1}{2} + \frac{1}{2}\cos 2\theta_{12}\cos 2\theta_{12}^m\Big) + s_{13}^4, \qquad \cos 2\theta_{12}^m = \frac{\cos 2\theta_{12} - A}{\sqrt{(\cos 2\theta_{12} - A)^2 + \sin^2 2\theta_{12}}}, \qquad A = \frac{2EV_{CC}c_{13}^2}{\Delta m^2_{21}}.$$

A resonance at the density $n_e = \Delta m^2\cos 2\theta/(2\sqrt{2}\,G_F E)$ is crossed adiabatically if the adiabaticity

$$\gamma = \frac{\Delta m^2 \sin^2 2\theta}{2E\cos 2\theta\,|d\ln n_e/dr|} \gg 1;$$

the probability of a jump between the matter eigenstates is the Landau-Zener expression $P_{LZ} = e^{-\pi\gamma/2}$.

9.3 The thermal potential

In the early Universe the electrons and positrons are relativistic and the potential is dominated by the thermal term of the W exchange (STANDARD; to leading order in T/m_W),

$$V_e - V_\mu = -\frac{8\sqrt{2}\,G_F p}{3m_W^2}\,\rho_{e^\pm}, \qquad \rho_{e^\pm} = \frac{7\pi^2}{60}T^4,$$

for a neutrino of momentum p; it suppresses the mixing as long as it exceeds $\Delta m^2\cos 2\theta/(2p)$.

9.4 The quantum kinetic equations and neutrino decoupling

In a gas of neutrinos the state of the neutrinos of momentum $\vec{p}$ is a 3×3 density matrix $\rho_{\vec{p}}$ (diagonal elements the occupation numbers of the flavours, off-diagonal ones the flavour coherence), and $\bar{\rho}_{\vec{p}}$ for antineutrinos. They obey (STANDARD)

$$i(\partial_{x_4} + \vec{v}_p\cdot\nabla)\rho_{\vec{p}} = \big[\Omega_{\vec{p}} + V + V_{\nu\nu}, \rho_{\vec{p}}\big] + i\,\mathcal{C}[\rho], \qquad \Omega_{\vec{p}} = \frac{U\,\mathrm{diag}(m_i^2)\,U^\dagger}{2p},$$

$$V_{\nu\nu} = \sqrt{2}\,G_F\int\frac{d^3q}{(2\pi)^3}\,(1 - \vec{v}_p\cdot\vec{v}_q)\big(\rho_{\vec{q}} - \bar{\rho}_{\vec{q}}\big), \qquad \vec{v}_p = \frac{\vec{p}}{p},$$

with $\mathcal{C}$ the collision term of $\mathcal{L}_{CC}$ and $\mathcal{L}_{NC}$ (in an expanding Universe $\partial_{x_4} \to \partial_{x_4} - Hp\,\partial_p$). During neutrino decoupling the energy exchanged between the plasma and the neutrinos is followed with one temperature per species (STANDARD; Escudero 2020): with the photon temperature T_γ, the neutrino temperatures T_{ν_e} and $T_{\nu_\mu} = T_{\nu_\tau}$, the Hubble rate $H = \sqrt{8\pi\rho/3}/M_{Pl}$ of the total density (here H is not the author's constant; $M_{Pl} = G_N^{-1/2} = 1.22091\times 10^{19}$ GeV), the energy densities $\rho_\gamma = \pi^2T_\gamma^4/15$ and $\rho_{\nu_\alpha} = 7\pi^2T_{\nu_\alpha}^4/120$ per flavour and the $e^\pm$ energy density ρ_e and pressure p_e (with the electron mass, $g = 4$, no chemical potential),

$$\frac{dT_\gamma}{dx_4} = -\frac{4H\rho_\gamma + 3H(\rho_e + p_e) + 3HT_\gamma\frac{dP_{int}}{dT_\gamma} + \frac{\delta\rho_{\nu_e}}{\delta x_4} + 2\frac{\delta\rho_{\nu_\mu}}{\delta x_4}}{\frac{d\rho_\gamma}{dT_\gamma} + \frac{d\rho_e}{dT_\gamma} + T_\gamma\frac{d^2P_{int}}{dT_\gamma^2}}, \qquad \frac{dT_{\nu_\alpha}}{dx_4} = -HT_{\nu_\alpha} + \frac{\delta\rho_{\nu_\alpha}/\delta x_4}{d\rho_{\nu_\alpha}/dT_{\nu_\alpha}},$$

$$\frac{\delta\rho_{\nu_e}}{\delta x_4} = \frac{G_F^2}{\pi^5}\Big[4(g_{eL}^2 + g_{eR}^2)F(T_\gamma, T_{\nu_e}) + 2F(T_{\nu_\mu}, T_{\nu_e})\Big], \qquad \frac{\delta\rho_{\nu_\mu}}{\delta x_4} = \frac{G_F^2}{\pi^5}\Big[4(g_{\mu L}^2 + g_{\mu R}^2)F(T_\gamma, T_{\nu_\mu}) - F(T_{\nu_\mu}, T_{\nu_e})\Big],$$

$$F(T_1, T_2) = 32f_a(T_1^9 - T_2^9) + 56f_sT_1^4T_2^4(T_1 - T_2), \qquad f_a = 0.884, \quad f_s = 0.829$$

(f_a, f_s the Fermi-Dirac factors of the annihilation and scattering rates; $g_{eL}=\frac12+s^2$, $g_{eR}=s^2$ for ν_e, which interacts with electrons through both currents, and $g_{\mu L}=-\frac12+s^2$, $g_{\mu R}=s^2$ for $\nu_{\mu,\tau}$, with $s^2=\sin^2\theta_W$, or the radiatively corrected values 0.727, 0.233, −0.273, 0.233). When oscillations equalise the flavours, one temperature T_ν obeys $dT_\nu/dx_4=-HT_\nu+(\delta\rho_{\nu_e}/\delta x_4+2\delta\rho_{\nu_\mu}/\delta x_4)/(3\,d\rho_\nu/dT_\nu)$. The finite-temperature QED corrections enter through the interaction pressure of the $e^\pm\gamma$ plasma, $P_{int}=P_2+P_3$, and its energy density $\rho_{int}=-P_{int}+T\,dP_{int}/dT$ (STANDARD; Bennett et al. 2020),

$$P_2=-\frac{e^2T^2}{12\pi^2}J_1-\frac{e^2}{8\pi^4}J_1^2+\frac{e^2m_e^2}{16\pi^4}J_2,\qquad P_3=\frac{e^3T}{12\pi^4}I^{3/2},$$

$$J_1=\int_0^\infty dp\,\frac{p^2}{E}n_D,\qquad J_2=\int_0^\infty\!\!\int_0^\infty dp\,dq\,\frac{pq}{E_pE_q}\ln\Big|\frac{p+q}{p-q}\Big|n_D(p)n_D(q),$$

$$I=\int_0^\infty dp\,\frac{p^2+E^2}{E}n_D,\qquad n_D=\frac{2}{e^{E/T}+1},\qquad E=\sqrt{p^2+m_e^2},$$

whose massless limits are $P_2=-5e^2T^4/288$ and $P_3=e^3T^4/(36\sqrt3\,\pi)$. The effective number of neutrino species is

$$N_{eff}=\Big(\frac{11}{4}\Big)^{4/3}\Big[\Big(\frac{T_{\nu_e}}{T_\gamma}\Big)^4+2\Big(\frac{T_{\nu_\mu}}{T_\gamma}\Big)^4\Big]$$

after e^+e^- annihilation. For instantaneous decoupling at T_d the plasma conserves its entropy a^3s, $s=(\rho_\gamma+\frac13\rho_\gamma+\rho_e+p_e+\rho_{int}+P_{int})/T$, while $T_\nu\propto1/a$.

9.5 The dense neutrino gas: the fast flavour instability

Near a supernova core $V_{\nu\nu}$ dominates $\Omega_{\vec p}$. For a homogeneous, axially symmetric gas of ν_e and $\bar\nu_e$ with the angular distributions $g_{\nu_e}(u)$, $g_{\bar\nu_e}(u)$ ($u=\cos\vartheta$ along the symmetry axis, $\int_{-1}^1 g\,du/2=1$), the electron-lepton-number (ELN) distribution

$$G(u)=\sqrt2\,G_F\big[n_{\nu_e}g_{\nu_e}(u)-n_{\bar\nu_e}g_{\bar\nu_e}(u)\big]$$

governs the linearised evolution of the small flavour coherence S_u (the ex element of ρ, x a heavy-lepton flavour). With $v=(1,u)$, $\Lambda^\mu=\int du/2\,G(u)\,v^\mu$ and modes $S_u\propto e^{-i(\Omega x_4-Kx_3)}$ (x_3 the symmetry axis) in the shifted variables $\omega=\Omega-\Lambda^0$, $k=K-\Lambda^3$, the linear equations have non-trivial solutions if (STANDARD)

$$\det\Pi(\omega,k)=0,\qquad \Pi^{\mu\nu}=\eta^{\mu\nu}+\int_{-1}^1\frac{du}{2}\,G(u)\,\frac{v^\mu v^\nu}{\omega-ku}\qquad(\eta=\mathrm{diag}(1,-1)\text{ on }(x_4,x_3)),$$

which splits into the axisymmetric branch and the azimuthal branch

$$(1+J_0)(-1+J_2)-J_1^2=0,\quad J_n=\int_{-1}^1\frac{du}{2}\,\frac{G(u)\,u^n}{\omega-ku};\qquad -1+\int_{-1}^1\frac{du}{2}\,\frac{G(u)\,(1-u^2)}{2(\omega-ku)}=0.$$

A root with $\mathrm{Im}\,\omega>0$ for real k is an instability with the growth rate $\mathrm{Im}\,\omega$. The homogeneous mode ($K=0$, i.e. $k=-\Lambda^3$) obeys

$$i\frac{dS_u}{dx_4}=(\Lambda^0-u\Lambda^3)\,S_u-\int_{-1}^1\frac{du'}{2}\,G(u')\,(1-uu')\,S_{u'},$$

and for the mode $K=\Lambda^3$ ($k=0$) the dispersion relation is the quadratic $\omega^2-(I_2-I_0)\omega-(I_0I_2-I_1^2)=0$, $I_n=\int du/2\,G\,u^n$, unstable with the growth rate $\frac12\sqrt{4I_1^2-(I_0+I_2)^2}$ when this is real. An instability exists only if $G(u)$ changes sign (an ELN crossing; Morinaga 2022).

9.6 The cosmic neutrino background

After decoupling the neutrinos keep the Fermi-Dirac distribution $f(p)=1/(e^{p/T_\nu}+1)$ of their momentum, redshifted with $T_\nu\propto1/a$, and lose their flavour coherence: the background is an incoherent mixture of the mass states, each with the number density and the energy density

$$n_i=2\int\frac{d^3p}{(2\pi)^3}f(p)=\frac{3\zeta(3)}{2\pi^2}T_\nu^3,\qquad \rho_i=2\int\frac{d^3p}{(2\pi)^3}\sqrt{p^2+m_i^2}\,f(p)$$

(the factor 2 counts ν_i and $\bar\nu_i$, one helicity each), and the mean velocity $\langle v\rangle_i=\int p^2\,(p/\sqrt{p^2+m_i^2})f\,dp/\int p^2 f\,dp$. The flavour content of the mass state ν_i is $|U_{\alpha i}|^2$.

10 The thermal history and the Boltzmann equations (version 3)

Degrees of freedom. In the hot Universe every species i of UM-2 in equilibrium (photon, gluons, W^+, W^-, Z, the six quarks, three charged leptons, three neutrinos, with $g_i = 2$, 16, 6 for W^+ and W^- together, 3, 12 per quark, 4 per charged lepton and 2 per neutrino; masses from the x_0 eigenvalues, the neutrinos massless; no Higgs particle; the right-handed neutrinos and the fold bands are not thermalised, and the dark matter is not counted, since it would add less than 0.1 to g_ρ at 5 TeV and nothing at its freeze-out) contributes the ideal-gas densities (g_i internal states, upper sign for fermions)

$$\rho_i = \frac{g_i}{2\pi^2}\int_0^\infty \frac{p^2 E\,dp}{e^{E/T}\pm 1}, \qquad p_i = \frac{g_i}{6\pi^2}\int_0^\infty \frac{p^4\,dp}{E\,(e^{E/T}\pm 1)}, \qquad s_i = \frac{\rho_i + p_i}{T},$$

$$g_\rho(T) = \frac{30}{\pi^2 T^4}\sum_i \rho_i, \qquad g_s(T) = \frac{45}{2\pi^2 T^3}\sum_i s_i, \qquad H = \Big(\frac{8\pi^3 g_\rho}{90}\Big)^{1/2}\frac{T^2}{M_{Pl}},$$

with the quark-gluon plasma weighted by $w(T) = \frac12[1+\tanh((T-T_c)/\Delta T)]$ and a gas of the lightest hadrons by $1-w(T)$ ($T_c = 155$ MeV, $\Delta T = 15$ MeV; ASSUMED model of the crossover; masses in GeV and internal states: π 0.1380, 3; K 0.4956, 4; η 0.5479, 1; ρ 0.7753, 9; ω 0.7827, 3; K^* 0.8917, 12; N 0.9389, 8; η' 0.9578, 1; ϕ 1.0195, 3; Λ 1.1157, 4; Σ 1.1932, 12; Δ 1.232, 32; from memory), and the neutrinos decoupled instantaneously at 2 MeV, after which their temperature follows from the separately conserved photon-electron entropy, $T_\nu/T = (g_{s,e\gamma}(T)/(11/2))^{1/3}$. The temperature falls as $d\ln T/dx_4 = -H/(1+\delta)$, $\delta = \frac13\, d\ln g_s/d\ln T$.

The dark matter. For one Dirac species of mass M ($Y = (n_\chi + n_{\bar\chi})/s$, $x = M/T$, $s = \frac{2\pi^2}{45}g_s T^3$)

$$\frac{dn_\chi}{dx_4} + 3Hn_\chi = -\langle\sigma v\rangle\big(n_\chi n_{\bar\chi} - n_{\chi,eq}^2\big) \quad\Longrightarrow\quad \frac{dY}{dx} = -\frac{(1+\delta)\,s\,\langle\sigma v\rangle}{2xH}\big(Y^2 - Y_{eq}^2\big), \qquad Y_{eq} = \frac{4M^2 T K_2(x)}{2\pi^2 s},$$

with the relativistic thermal average of the total annihilation cross-section (Gondolo and Gelmini)

$$\langle\sigma v\rangle = \frac{1}{8M^4 T K_2(x)^2}\int_{4M^2}^\infty \sigma(s)\,(s-4M^2)\,\sqrt{s}\,K_1(\sqrt{s}/T)\,ds,$$

and the relic $\Omega h^2 = MY_\infty s_0/(\rho_c/h^2)$, $\rho_c/h^2 = 1.05368\times10^{-5}$ GeV/cm^3, s_0 today. The cross-sections are (spin averaged; $\beta = \sqrt{1-4M^2/s}$, $p = \sqrt{s/4 - M^2}$ the centre-of-mass momentum)

$$\sigma_{I12} = \frac{G^2 s^{3/2}(s-4M^2)^{1/2}}{8\pi}, \quad G^2 = 8G_g^2 + 3G_W^2 + G_B^2; \qquad \sigma_{I13} = \sum_f \frac{N_c G_f^2\, s\,(1+2M^2/s)}{24\pi\beta}, \quad G_f = G_A\,\tfrac13\,\tilde I_f^{(b)},$$

the first from $\sigma v_{rel} = G^2 s(s-4M^2)/(4\pi)$ per abelian channel with the relative velocity $v_{rel} = 2\beta$ in the centre-of-mass frame (p-wave, from ${}^3P_0$ only), the second summed over the quark Weyl components (s-wave, from ${}^3S_1$), and partial-wave unitarity bounds them by

$$\sigma_{I12} \le \frac{\pi}{4p^2} \quad (J=0), \qquad \sigma_{I13} \le \frac{3\pi}{4p^2} \quad (J=1);$$

where the effective cross-section exceeds its bound it is replaced by the bound (a unitarity-respecting completion that saturates it).

The fold bands (freeze-in through I11). With $G_f = G_V q_f \tilde I_f$ and the fold-band overlaps $\tilde I_f$ (the sum below runs over the fifteen Standard-Model multiplets of three generations in the plasma, each with its N_c Weyl components; the ν_R are not in the plasma), the pair-production rate density and the energy density of the fold bands obey (Maxwell-Boltzmann, massless)

$$\gamma = \frac{T}{32\pi^4}\int_0^\infty ds\,\Big(\sum_{\text{spins}}\sigma\Big)\,s^{3/2}K_1(\sqrt{s}/T) = \frac{4}{\pi^5}N_\chi G_V^2 T^8 \sum_f N_c(q_f\tilde I_f)^2, \qquad \frac{d(\rho_{fold}a^4)}{dx_4} = 8T\,\gamma\,a^4,$$

($\int_0^\infty x^6 K_1(x)dx = 384$, $\int_0^\infty x^7 K_2(x)dx = 3072$; $a \propto g_s^{-1/3}/T$), integrated from the highest temperature T_{max} of the hot Universe; today $\Delta N_{eff} = \rho_{fold}/\rho_\nu$ with $\rho_\nu = \frac78\frac{\pi^2}{15}T_\nu^4$ per flavour. They stay out of equilibrium if $2\gamma/(n_{fold,eq}H) \ll 1$, $n_{fold,eq} = N_\chi\cdot 4\cdot\frac34\frac{\zeta(3)}{\pi^2}T^3$.

The right-handed neutrinos. If I13 coupled to the $B-L$ current instead of J^B (charge -1 for the leptons and the ν_R), the relic would need the strength $G_A^{B-L} = G_A\,[S_B/S_{B-L}]^{1/2}$, since the effective cross-section scales with $S = \sum_f N_c (q_f \tilde{I}_f^{(b)})^2$ (S_B over the quarks, S_{B-L} over the quarks, the leptons and the ν_R; G_A here the value without the unitarity cap), and one flavour ν_R would couple to the dark matter with $G = G_A^{B-L}\tilde{I}_{\nu_R}^{(b)}$. The number density of ν_R particles would then obey

$$\frac{dn_{\nu_R}}{dx_4} + 3Hn_{\nu_R} = \gamma_\nu\Big(1 - \frac{n_{\nu_R}^2}{n_{\nu_R,eq}^2}\Big), \qquad \gamma_\nu = \sum_{A,B} n_{\chi,eq} n_{\bar\chi,eq} \frac{G^2M^2}{2\pi}, \qquad n_{\nu_R,eq} = \frac{3}{4}\frac{\zeta(3)}{\pi^2}T^3,$$

($n_{\chi,eq} = n_{\bar\chi,eq} = 2(MT/2\pi)^{3/2}e^{-x}$ per Dirac species; $G^2M^2/(2\pi)$ is the s-wave σv into one Weyl ν_R pair), while elastic scattering on the dark matter keeps the ν_R at the plasma temperature as long as

$$\Gamma_{el} = \sum_{A,B} (n_\chi + n_{\bar\chi})\,\sigma_{el} > H, \qquad \sigma_{el} = \frac{G^2E^2}{2\pi}$$

for a Weyl ν_R of energy $E \simeq 3T$ on a heavy Dirac fermion (ESTIMATE); a species of n_W Weyl fermions in equilibrium until the plasma has g_s entropy degrees of freedom adds $\Delta N_{eff} = n_W(10.75/g_s)^{4/3}F_h$. With J^B the ν_R are produced only through I11, with $G = G_V\tilde{I}_{\nu_R}$ (the overlap of the dark-matter orbital with the ν_R profile), at the fraction $G_V^2\sum_i \tilde{I}_{\nu_R i}^2 \big/ \big(G_A^2\sum_f N_c(q_f\tilde{I}_f^{(b)})^2\big)$ of the I13 annihilation rate into quarks.

The self-interaction channel. The contact term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ contains, for the dark-matter orbital (a_D, b_D) and the fold zero mode $(a_F, 0)$ ($b_F = 0$), the four-dimensional term $-\lambda(\bar\chi_D\chi_F)(\bar\chi_F\chi_D)\,s_{DF}(y)^2$ with the transition scalar density $s_{DF} = a_Db_F + b_Da_F$, which is odd under the fold reflection while its square is even; so the dark matter annihilates into fold pairs with

$$G_\lambda = \frac{\lambda H\tilde{I}_\lambda}{2f^4v_t}, \qquad \tilde{I}_\lambda = \int_{-L}^{0} s_{DF}^2\,e^{-3Hy}\,dy \ \text{ (units of } H), \qquad \sigma v = \frac{G_\lambda^2M^2}{2\pi} \text{ per final fold species (ESTIMATE),}$$

and while the dark matter is in equilibrium each annihilation puts $2M$ into the fold bands:

$$\frac{d(\rho_{fold}a^4)}{dx_4} = 2M\,\gamma_\lambda\,a^4, \qquad \gamma_\lambda = \sum_{A,B} n_{\chi,eq}n_{\bar\chi,eq}\,\sigma v\,N_F,$$

integrated from T_{max} to the freeze-out. The fold bands stay out of equilibrium if $2\gamma_\lambda/(n_{fold,eq}H) \ll 1$; otherwise they decouple as four Weyl species, $\Delta N_{eff} = 4(10.75/g_s)^{4/3}F_h$, at the lowest temperature at which $2\gamma_\lambda/(n_{fold,eq}H) \geq 1$.

11 The coupled system and how it is solved

The fields couple as follows: gravity to everything (the canonical spin connection for Ψ and the fermions, the metric in every kinetic term, the stabiliser); Ψ to gravity, to itself (λ), to the fermion currents (I11), to the gauge-field invariants (I12) and, with its x_0-chiral current, to the baryon-number current (I13); the fermions of the three generations to the gauge fields, to Σ at the tip (through the tip matrices, which carry the flavour mixing) and to Ψ (I11, and I13 for the quarks); the gauge fields to the fermions, to Σ at the tip and to Ψ through I12 (a field-dependent coupling). UM-2 solves the coupled system in the following hierarchy, each step feeding the next:

1. the background: the static block $(A, h, k) = (0, 0, k_*)$ with the stabiliser, its stability and its first-order response to every block-averaged source; the primordial approach to it from the full nonlinear equations;
2. the x_0 eigenvalue problems on the static block: all masses and the mixing matrices (gauge bosons, graviton, the twelve fermions with their tip matrices, Ψ);
3. the Kohn-Sham equations of Ψ with the portal mean fields of the other fields and the self-interaction; the Kohn-Sham energy-momentum tensor returned to step 1 (the adiabatic block response);
4. the flavour evolution of the neutrinos (section 9) in vacuum, in matter, in the early Universe (decoupling and N_{eff}, which feed step 5) and in dense neutrino gases;
5. the four-dimensional effective theory and its linear perturbations (CMB, CNB, BAO);
6. (version 3) the coupled iteration in environments: the Kohn-Sham levels with the mean fields of section 7.1 from the densities and condensates of the environment, the field-dependent couplings and the W, Z eigenvalues with the dark-matter density, and the full adiabatic block response to the trace $\rho - 3p_3$ and the hidden pressure $\bar p$ of all species (the plasma included), repeated until the masses are self-consistent;

7. (version 3) the thermal history: the degrees of freedom from the x_0 masses, the Boltzmann equations of section 10 for the dark matter (relic), the fold bands (freeze-in and the self-interaction channel) and the right-handed neutrinos, and the terms of the metric series beyond $n = 0$ (section 8.1).

12 What was verified

- the Lovelock tensors of the order-0 metric by four routes, the Bianchi identities ($k = 1, 2, 3$), the record's equations and the minisuperspace variation (sympy, exact);
- the static block, the stabiliser, the radion, the stabiliser family, the first-order response and the 4D Friedmann acceleration equation, the dimensional reduction (sympy, exact);
- the exact Dirac reduction in all eight blocks, the boundary conditions, the zero modes, the tip mixing condition of one generation and of three generations for an arbitrary complex tip matrix, the gauge x_0 operator and the gauge invariance of the tip terms with arbitrary generation matrices;
- the x_0 spectra (mpmath; finite elements to 5×10^{-10}; shooting to 6×10^{-10}); the three-generation tip determinant (mpmath, 50 digits: its roots equal the one-generation roots to 3.0×10^{-42}, its null vectors the columns of the tip matrix); the absence of flavour-changing Kaluza-Klein couplings (10^{-17});
- the gauge anomalies of the three generations (exact fractions);
- the flavour evolution (exact evolution operators against the vacuum formula to 3.9×10^{-15}, unitarity, CPT, the T-odd difference; the numerical solar evolution against the adiabatic formula to 1.3×10^{-3}; the exact 1-3 resonance against its formula to 1.6×10^{-4}; the massless limits of the QED integrals; the decoupling equations against the published values of the same method; three independent evaluations of the fast-flavour growth rates to 1.2×10^{-14});
- the Kohn-Sham solver (record reference states to 10^{-13}; energy identity; conservation law; Richardson; Hellmann- Feynman for every portal mean field);
- (version 3) the overlaps $\tilde{I}$ and $\tilde{I}^{(b)}$ against the derivatives of self-consistent Kohn-Sham levels (to 3×10^{-10}), the vanishing of I13 on the fold bands (to 7×10^{-14}), the screening of the response in the interacting state, the fold band massless under I11; the Bessel moments of the freeze-in rate; the Boltzmann solver against the late-time integral (1.3×10^{-8}) and the freeze-out formula; the thermal average against its non-relativistic limit; the degrees of freedom against their exact limits (105.75, 86.25, 10.75, 43/11); the graviton tower against finite elements (4×10^{-9}) and the orthogonality of the flat profile (2×10^{-10}); the potential from the Weyl variable against the Poisson equation (1.5×10^{-6});
- (version 3, after the independent review) the algebra of the I13 bilinear with the author's matrices (Hermitian for $M \neq x_0$, not for $M = x_0$; its B structure; exact); the cancellation of radiation in the block response ($c_3 = -3c_\rho$); the Feynman-Hellmann pressure against the trace-anomaly gluon condensate (10^{-16}); the relativistic rate fractions above the unitarity crossings by adaptive quadrature with the breakpoints of the thermal weight.